immunity and salmonella constructs may give oral and systemic immunity. Intranasal immunization is fast gaining popularity.

• The risk with live attenuated organisms is reversion to the virulent form and danger to immunocompromised individuals.

Subunit vaccines

• Whole organisms have a multiplicity of antigens, some of which are not protective, may induce hypersensitivity or might even be frankly immunosuppressive.

• It makes sense in these cases to use purified components.

• There is greatly increased use of recombinant DNA technology to produce these antigens. Expression in bananas provides a very cheap way of achieving oral immunization in the developing world.

• Naked DNA encoding the vaccine subunit can be injected directly into muscle, where it expresses the protein and produces immune responses. The advantages are stability, ease of production and cheapness.

• Epitope-specific vaccines based on conserved structures have the advantage that they can provide broad protection and may avoid the possible deleterious effects of other epitopes (autoimmunity, T-downregulation, original antigenic sin, escape by mutation of immunodominant epitopes) when certain whole antigens are used for immunization.

• Epitope-specific vaccines can be based on peptides, internal image anti-idiotypes or epitope-loss mutants.

• Peptides may only usefully mimic the native protein for vaccination to produce antibody if the epitope is linear and relatively unconstrained in structure. Carriers such as tetanus toxoid or mycobacterial heat-shock proteins are needed to make the peptide immunogenic. Linear peptides can mimic T-cell epitopes in the whole protein.

• Epitope-loss mutants have undesirable epitopes replaced but still fold correctly to produce the wanted discontinuous B-cell epitope(s).

Current vaccines

• Children in the USA and UK are routinely immunized with diphtheria and tetanus toxoids and pertussis (DTP triple vaccine) and attenuated strains of measles, mumps and rubella (MMR) and polio. BCG is given at 10–14 years.

• Subunit forms of pertussis lacking side-effects are being introduced.

• The capsular polysaccharide of *H. influenzae* has to be linked to a carrier.

• The elderly receive vaccines of influenza and *Pneumococcus* polysaccharides.

• Vaccines for hepatitis A and B, meningitis, yellow fever, typhoid, cholera and rabies are available for travelers and high-risk groups.

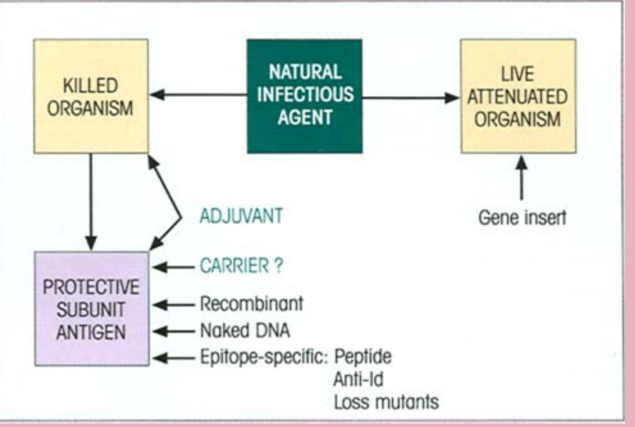

Figure 14.18. Strategies for vaccination.

Vaccines in development

• In malaria, the experimental vaccines are targeted at the *sporozoite and blood stages*, the *toxin* producing serious side-effects, the *gametes* and the insect gut trypsin.

• Recombinant glutathione-*S*-transferase is a promising candidate for a vaccine against schistosomiasis.

• An oral vaccine composed of cholera toxin B subunit and killed vibrios induced good mucosal immunity to cholera.

Adjuvants

• Adjuvants work by producing depots of antigen, and by activating macrophages; they sometimes have direct effects on lymphocytes.

• Adjuvants such as the muramyl dipeptide analogs derived from mycobacterial cell walls and the monophosphoryl lipid A derivative from Gram-negative LPS may soon be in general use.

• New methods of delivery include linking the antigen to small lipid membrane vesicles (liposomes) or a special glycoside matrix (Iscom). These delivery particles can be furnished with many factors which improve their immunogenicity and flexibility. One can build in several antigens into the same particle, adjuvants such as MDP and MLA, cytokines to influence lymphocyte subset responses and molecules such as cholera toxin B and *E. coli* enterotoxin to target particular sites in the body.

• Antigens built into biodegradable polymers of varying half-life can provide single-shot vaccines which mimic a conventional course of immunization requiring several injections.

The overall strategies for vaccination are summarized in figure 14.18.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

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Immunodeficiency

INTRODUCTION

In accord with the dictum that 'most things that can go wrong, do so', a multiplicity of immunodeficiency states in humans, which are **not secondary** to environmental factors, have been recognized. These 'experiments of nature' provide valuable clues regarding the function of the defective factors concerned. We have earlier stressed the manner in which the interplay of complement, antibody and phagocytic cells constitutes the basis of a tripartite defense mechanism against pyogenic (pus-forming) infections with bacteria which require prior opsonization before phagocytosis. It is not surprising, then, that deficiency in any one of these factors may predispose the individual to repeated infections of this type. Patients with T-cell deficiency of course present a markedly different pattern of infection, being susceptible to those viruses and moulds which are normally eradicated by cell-mediated immunity (CMI).

A relatively high incidence of malignancies, and of autoantibodies with or without autoimmune disease, has been documented in patients with immunodeficiency possibly due to failure of T-cell regulation or inability to control key viral infections.

The following sections examine various forms of these **primary immunodeficiencies** which have a genetic basis.

DEFICIENCIES OF INNATE IMMUNE MECHANISMS

Phagocytic cell defects

In **chronic granulomatous disease** the monocytes and polymorphs fail to produce reactive oxygen intermediates (figure 15.1) due to a defect in the cytochrome b_{558} oxidase system (cf. p. 4) normally activated by phagocytosis. The cytochrome has 92 and 22 kDa subunits and, in the X-linked form of the disease, there are mutations in the gene encoding the larger of these subunits. In the majority of cases, no cytochrome is produced, but one variant gp92 mutation permits the synthesis of low levels of the protein (figure 15.1) and the condition can be improved by treatment with γ-interferon. The 30% of chronic granulomatous disease patients who inherit their disorder in an autosomal recessive pattern express a defective form of the oxidase resulting from mutations in the smaller $p22^{phox}$ cy-

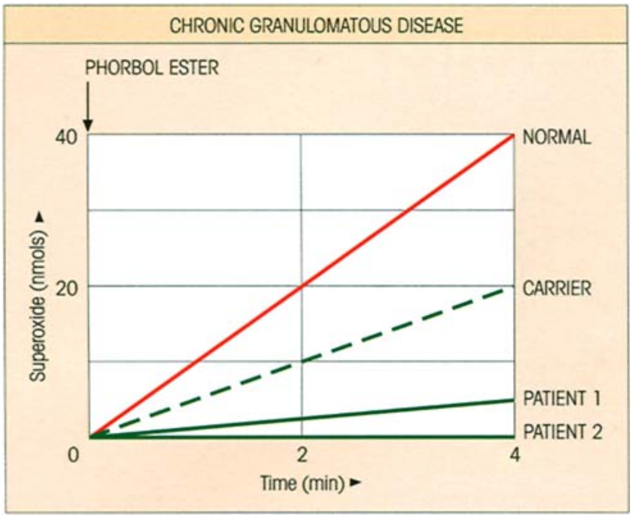


Figure 15.1. Defective respiratory burst in neutrophils of patients with chronic granulomatous disease. The activation of the NADP/cytochrome oxidase is measured by superoxide anion ($\cdot O_2^-$; cf. figure 1.10) production following stimulation with phorbol myristate acetate. Patient 2 has a p92^{phox} mutation which prevents expression of the protein, whilst patient 1 has the variant p92^{phox} mutation producing very low but measurable levels. Many carriers of the X-linked disease express intermediate levels, as in the individual shown who is the mother of patient 2. (Data from Smith R.M. & Curnutte J.T. (1991) *Blood* **77**, 673.)

tochrome subunit and in the cytosolic p47^{phox} and p67^{phox} (*phox* = phagocyte oxidase) components of the total reduced nicotinamide adenine dinucleotide phosphate (NADPH) oxidase system. Not unexpectedly, the knockout of gp92^{phox} provides a handy mouse model (cf. figure 1.10).

Curiously, the range of infectious pathogens which trouble these patients is relatively restricted. The most common pathogen is *Staphylococcus aureus*, but certain Gram-negative bacilli and fungi such as *Aspergillus fumigatus* and *Candida albicans* are frequently involved. The factors underlying this restriction are two-fold. First, many bacteria help to bring about their own destruction by generating H_2O_2 through their own metabolic processes, but if they are catalase positive, the peroxide is destroyed and the bacteria will survive. Thus polymorphs from these patients readily take up catalase-positive staphylococci in the presence of antibody and complement but fail to kill them intracellularly. Second, the organisms which are most virulent tend to be those that are highly resistant to the oxygen-independent microbicidal mechanisms of the phagocyte.

Lack of the CD18 β subunit of the β_2 integrins produces a **leukocyte adhesion deficiency** causing impaired neutrophil chemotaxis and recurrent bacterial infection. Emigration of monocytes, eosinophils and lymphocytes is unaffected since these can fall back on the alternative VCAM-1/VLA-4 β_1 integrin system. Two cases (remember these primary immunodeficiencies tend to be pretty rare) of leukocyte adhesion deficiency associated with mental retardation and Bombay hh blood group phenotype had defective neutrophil motility due to a failure to produce sialyl Lewis^x, the ligand used to bind to the selectins on endothelial cells. The defect in the synthesis of blood group H antigen, which is also a fucosylated carbohydrate, suggests a defect in fucose metabolism. In the not too distant future, these deficiencies will be treated by gene replacement but, in the meantime, allogeneic bone marrow grafts used to correct the CD18-linked disease are surprisingly well tolerated, possibly because of the LFA-1 defect.

In **Chediak–Higashi** disease and the 'beige' murine counterpart, dysfunction of NK cells and neutrophils is associated with the accumulation of giant intracytoplasmic granules which result from defective trafficking of the late endosomal/lysosomal compartment. The patients suffer from sometimes fatal pyogenic infections, particularly with *Staphylococcus aureus*. Infection often triggers a mysterious 'accelerated phase' of unremitting T-cell proliferation, but this can be brought under control by bone marrow transplantation.

Mendelian susceptibility to mycobacterial infection in humans involving BCG or nontuberculous mycobacteria can be traced to recessive mutations in four genes, two affecting the chains of the IFNγ receptor (IFNγR1 and IFNγR2) and, the others, the IL-12 p40 and IL-12R β1 subunits. Apart from some increased susceptibility to *Salmonella*, no other opportunistic infections were observed, highlighting the selectivity of the IFNγ protective role. To put things in perspective, mycobacteria induce IL-12 secretion by macrophages and dendritic cells which then activates both NK and T-cells; these then produce IFNγ which turns on the macrophage mechanisms providing protection against less pathogenic mycobacteria.

There is a group of so-called **'autoinflammatory' disorders** characterized by apparently unprovoked inflammation unrelated to high titer autoantibodies or antigen-specific T-cells. One such, **TNF receptor-associated periodic syndrome** (TRAPS), presents with prolonged fever bouts and severe localized inflammation caused by dominantly inherited mutations in the *TNFRSFIA* gene encoding the 55 kDa TNF receptor. Impaired cleavage of the receptor ectodomain with diminished shedding of the potentially antagonistic soluble receptor may account for the persistent inflammatory signal, and it is salutary to note the beneficial effects of treatment with a recombinant p75 TNFR-Fcγ fusion protein or anti-TNF. Yet another hereditary periodic fever syndrome is **familial Mediterranean fever** due to mutations in the *MEFV* gene that encodes a 95 kDa inflammatory regulator denoted as pyrin or marenostrin, expressed predominantly in neutrophils and Th1 cytokine-activated monocytes.

Complement system deficiencies

Defects in control proteins

The importance of complement in defense against infections is emphasized by the occurrence of repeated life-threatening infection with pyogenic bacteria in patients lacking factor I, the C3b inactivator. Because of this inability to destroy C3b, there is continual activation of the alternative pathway through the feedback loop, leading to very low C3 and factor B levels with normal C1, C4 and C2.

Each red blood cell is bombarded daily with 1000 molecules of C3b generated through the formation of fluid phase alternative pathway C3 convertase from the spontaneous hydrolysis of the internal thiolester of C3. There are several regulatory components on the red cell surface to deal with this. The C3 convertase complex is dissociated by decay accelerating factor (DAF; CD55) and by CR1 complement receptors (not forgetting factor H from the fluid phase; cf. p. 11), after which the C3b is dismembered by factor I in concert with CR1, membrane cofactor protein (MCP) or factor H (figure 15.2). There are also two inhibitors of the membrane attack complex, homologous restriction factor (HRF) and the abundant protectin molecule (CD59) which, by binding to C8, prevent the unfolding of the first C9 molecule needed for membrane insertion. DAF, HRF and CD59 bind to the membrane through glycosyl phosphatidylinositol anchors. In a condition known as **paroxysmal nocturnal hemoglobinuria (PNH)**, there is a defect in the ability to synthesize these anchors, caused by a mutation in the X-linked *PIG-A* gene encoding the enzyme required for adding *N*-acetyl glucosamine to phosphatidylinositol. In the absence of these complement regulators, serious lysis of the red cells occurs. In the less severe type II PNH, there is a defect in DAF, but in the type III form, associated also with deficiency of CD59, susceptibility to spontaneous complement-mediated lysis is greatly increased (figure 15.2). The erythrocytes can be normalized by adding back the deficient factors.

In **acute myocardial infarction**, the expression of the CD59 protectin decreases, perhaps due to the shedding of small membrane vesicles, and this sensitizes the injured myocardial cells to attack by the membrane attack complex (MAC) leading to a clear demarcation between nonviable and viable tissue areas.

An inhibitor of active C1 is grossly lacking in **hereditary angioedema** and this can lead to recurring episodes of acute circumscribed noninflammatory

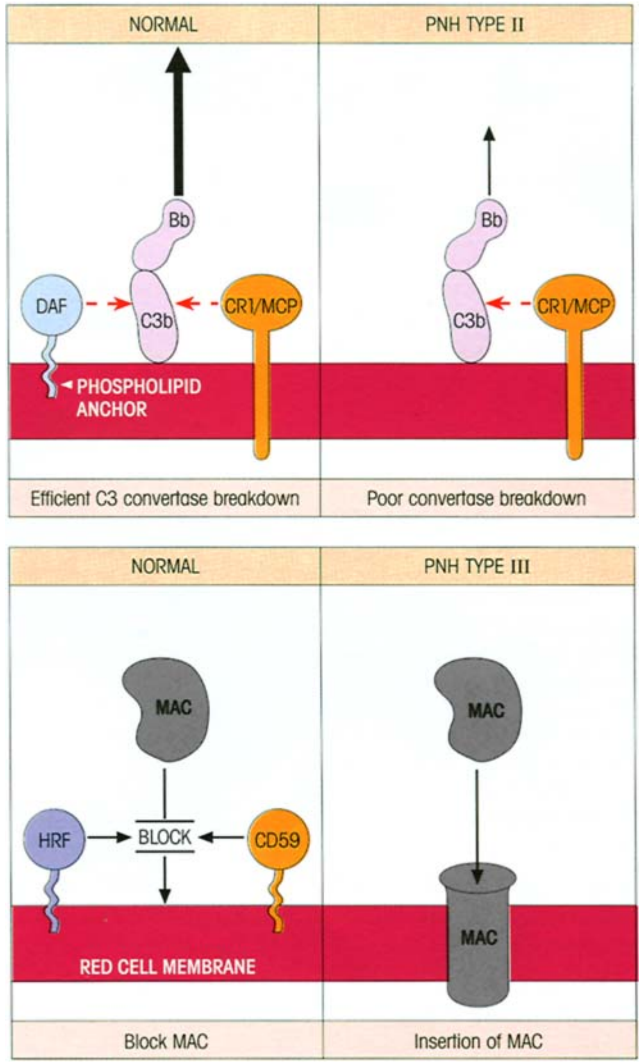


Figure 15.2. Paroxysmal nocturnal hemoglobinuria (PNH). A mutation in the *PIG-A* gene, which encodes α-1,6-*N*-acetyl glucosaminyl-transferase, results in an inability to synthesize the glycosyl phosphatidylinositol anchors, deprives the red cell membrane of complement control proteins and renders the cell susceptible to complement-mediated lysis. Type II is associated with a DAF defect and the more severe type III with additional CD59 deficiency. DAF, decay accelerating factor; CR1, complement receptor type 1; MCP, membrane cofactor protein; HRF, homologous restriction factor; MAC, membrane attack complex.

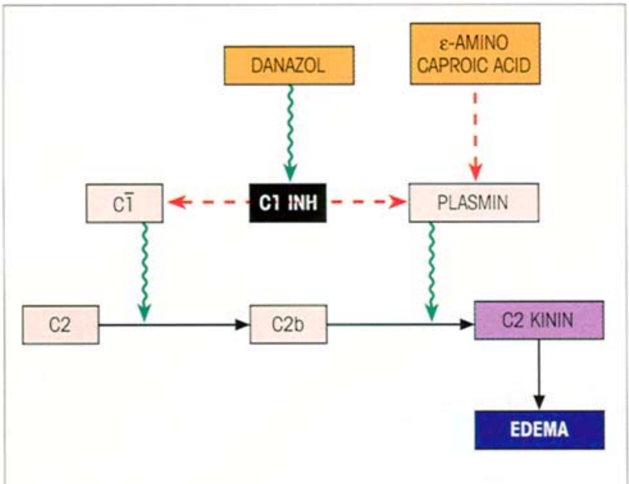


Figure 15.3. C1 inhibitor deficiency and angioedema. C1 inhibitor stoichiometrically inhibits C1, plasmin, kallikrein and activated Hageman factor and deficiency leads to formation of the vasoactive C2 kinin by the mechanism shown. The synthesis of C1 inhibitor can be boosted by methyltestosterone or preferably the less masculinizing synthetic steroid, danazol; alternatively, attacks can be controlled by giving ε-aminocaproic acid to inhibit the plasmin.

edema mediated by a vasoactive C2 fragment (figure 15.3). The patients are heterozygotes and synthesize small amounts of the inhibitor which can be raised to useful levels by administration of the synthetic anabolic steroid danazol or, in critical cases, of the purified inhibitor itself. ε-Aminocaproic acid, which blocks the plasmin-induced liberation of the C2 kinin, provides an alternative treatment.

Deficiency of components of the complement pathway

Deficiencies in C1q, C1r, C1s, C4 and C2 predispose to the development of immune complex diseases such as SLE (cf. p. 425), perhaps due to a decreased ability to mount an adequate host response to infection with a putative etiologic agent or, more probably, to eliminate antigen–antibody complexes effectively (cf. p. 337). Bearing in mind the focus of the autoimmune response in SLE on the molecular constituents of the blebs appearing on the surface of apoptotic cells (cf. p. 427), the importance of C1q in binding to and clearing these apoptotic bodies becomes paramount. So it is that C1q-deficient mice develop high titer antinuclear antibodies and die with severe glomerulonephritis. In another twist to the story, apoptotic bodies coated with C1q and C4 migrate to the bone marrow where they bind the C4b receptor and negatively select self-reactive B-cells, as evidenced by the activation of antinuclear B-cells in C4-deficient animals.

Permanent deficiencies in C5, C6, C7, C8 and C9 have all been described in humans, yet in virtually every case the individuals are healthy and not particularly prone to infection, apart from an increased susceptibility to disseminated *Neisseria gonorrhoeae* and *N. meningitidis*. Thus full operation of the terminal complement system does not appear to be essential for survival, and adequate protection must be largely afforded by opsonizing antibodies and the immune adherence mechanism.

PRIMARY B-CELL DEFICIENCY

X-Linked (Bruton-type) a-γ-globulinemia is due to early B-cell maturation failure

Bruton's congenital a-γ-globulinemia is one of several immunodeficient syndromes which have been mapped to the X chromosome (figure 15.4). The defect occurs at the pre-B-cell stage and the production of immunoglobulin in affected males is grossly depressed, there being few lymphoid follicles or plasma cells in lymph node biopsies. Mutations occur in Bruton's tyrosine kinase (*Btk*) gene, which is also responsible for the *xid* defect in mice. The children are subject to repeated infection by pyogenic bacteria—*Staphylococcus aureus*, *Streptococcus pyogenes* and *S. pneumoniae*, *Neisseria meningitidis*, *Haemophilus influenzae*—and by a rare protozoan, *Pneumocystis carinii*, which produces a strange form of pneumonia. Cell-mediated immune responses are normal and viral infections such as measles and smallpox are readily brought under control. Therapy involves the repeated administration of human γ-globulin to maintain adequate concentrations of circulating immunoglobulin.

Mutations in either the μ heavy chain or the λ_5 chain, which contribute to the surrogate IgM receptor on pre-B-cells (cf. p. 238), result in a phenotype similar to that seen in Bruton's a-γ-globulinemia with arrest at the pro-B stage. The implication would be that *Btk* provides the signal for pro- to pre-B differentiation through this pre-B-cell receptor complex.

IgA deficiency and common variable immunodeficiency (CVID) have a similar genetic basis

These diseases, which are the first and second most common primary immunodeficiencies, represent the extreme ends of a spectrum of immunoglobulin defi-

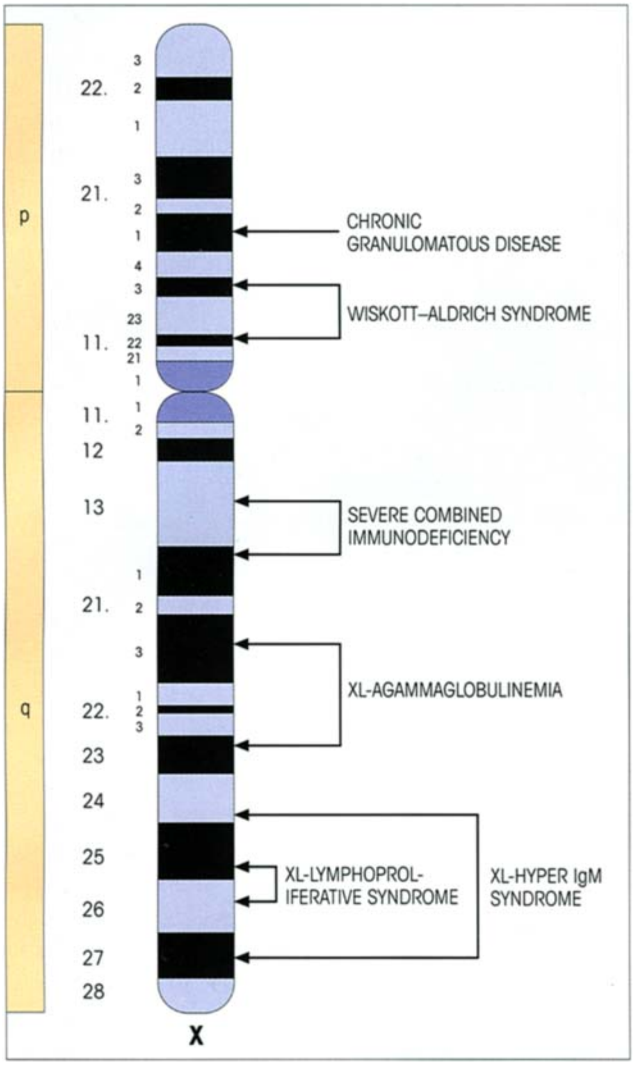


Figure 15.4. Loci of the X-linked (XL) immunodeficiency syndromes. Males are more likely to be affected by X-linked recessive genes because, unlike the situation with females where there are two X chromosomes, homozygosity is not needed.

ciencies. IgA deficiency, which may include IgG2, and CVID often occur within the same family with frequent sharing of HLA haplotype, and individual family members may gradually convert from one disease to the other. This suggests a common genetic basis, but the heterogeneity of the disease complicates the search for candidate genes which has not so far progressed beyond the identification of one or more susceptibility loci within the TNF/lymphotoxin region of the MHC class III genes. Likewise, the cellular defects have not been satisfactorily pinpointed: the majority of patients have a defect in T-cell priming by antigen which *might* be attributable to an antigen-presenting cell dysfunction, and most patients reveal a pattern of raised CD8 IFNγ production, increased expression of HLA-DR and Fas by CD4 cells and an increased rate of apoptosis. CVID patients have to be protected against recurrent pyogenic infections with intravenous γ-globulin but, strangely, HIV infection restores the production of IgG and IgM, but not IgA, antibodies. This could mean that CVID is potentially reversible by immunoregulatory factors and lends support to the notion that selective IgA deficiency predisposes to CVID.

Transient hypo-γ-globulinemia is seen in early life

A degree of immunoglobulin deficiency occurs naturally in human infants as the maternal IgG level wanes, and may become a serious problem in very premature babies. A more protracted **transient hypo-γ-globulinemia of infancy**, characterized by recurrent respiratory infections, is associated with low IgG levels which often return somewhat abruptly to normal by 4 years of age. There is a deficiency in the number of circulating lymphocytes and in their ability to generate help for Ig production by B-cells activated by pokeweed mitogen, but this becomes normal as the disease resolves spontaneously.

PRIMARY T-CELL DEFICIENCY

Patients with no T-cells or poor T-cell function are vulnerable to opportunistic infections and, since B-cell function is to a large extent T-dependent, T-cell deficiency also impacts negatively on humoral immunity. Dysfunctional T-cells often permit the emergence of allergies, lymphoid malignancies and autoimmune syndromes, the latter presumably arising from inefficient negative selection in the thymus or the failure to generate appropriate regulatory cells.

Some deficiencies affect early T-cell differentiation

The **DiGeorge** and **Nezelof syndromes** are characterized by a failure of the thymus to develop properly from the third and fourth pharyngeal pouches during embryogenesis (DiGeorge children also lack parathyroids and have severe cardiovascular abnormalities). Consequently, stem cells cannot differentiate to become T-lymphocytes and the ‘thymus-dependent’ areas in lymphoid tissue are sparsely populated; in contrast, lymphoid follicles are seen but even these are poorly developed (figure 15.5). Cell-mediated immune responses are undetectable and, although the infants can deal with common bacterial infections, they

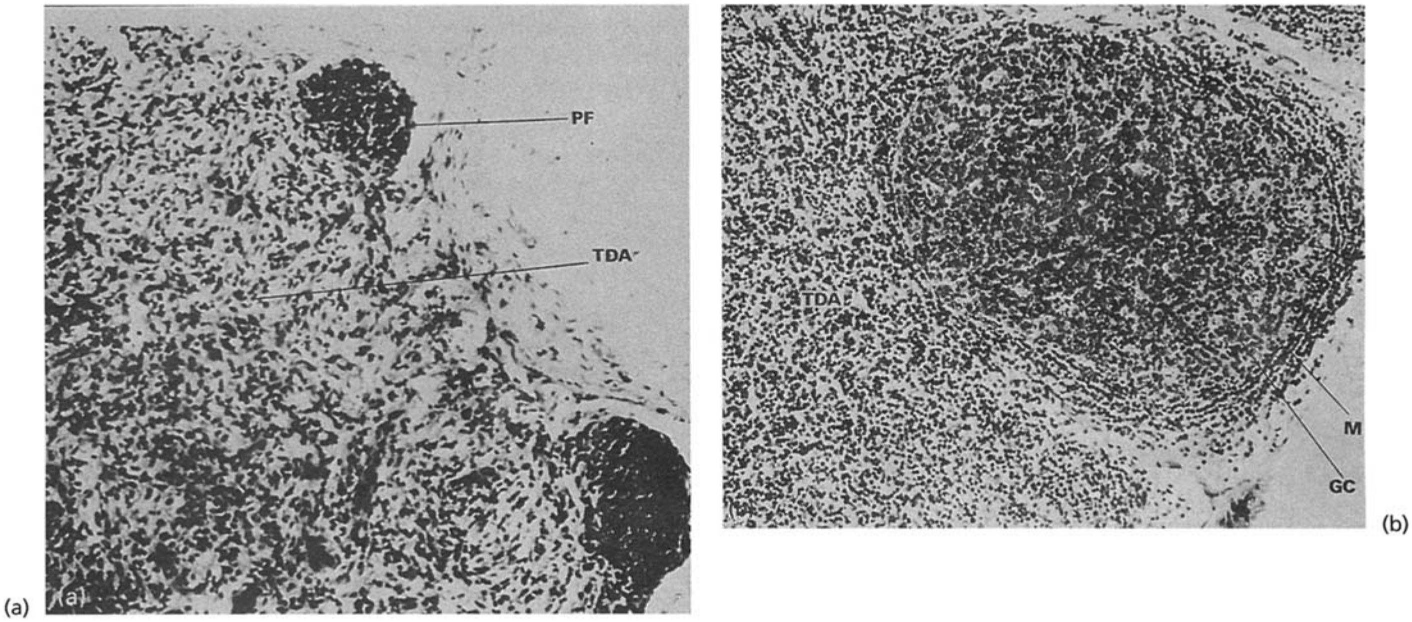


Figure 15.5. Lymph node cortex. (a) From patient with DiGeorge syndrome showing depleted thymus-dependent area (TDA) and small primary follicles (PF); (b) from normal subject: the populated T-cell area and the well-developed secondary follicle with its mantle of small lymphocytes (M) and pale-staining germinal center (GC) provide a marked contrast. (DiGeorge material kindly supplied by Dr D. Webster; photograph by Mr C.J. Sym.) In the murine model, the athymic nude mouse, there is an abnormality in the winged helix protein.

may be overwhelmed by vaccinia (figure 15.6) or measles, or by bacille Calmette–Guérin (BCG) if given by mistake. Humoral antibodies can be elicited, but the response is subnormal, presumably reflecting the need for the cooperative involvement of T-cells. (The similarity of this condition to neonatal thymectomy and of B-cell deficiency to neonatal bursectomy in the chicken should not go unmentioned.) Treatment by grafting neonatal thymus leads to restoration of immunocompetence, but some matching between the major histocompatibility antigens on the nonlymphocytic thymus cells and peripheral cells is essential for the proper functioning of the T-lymphocytes (p. 228). Complete absence of the thymus is pretty rare and more often one is dealing with a 'partial DiGeorge' in which the T-cells may rise from 6% at birth to around 30% of the total circulating lymphocytes by the end of the first year; antibody responses are adequate.

Mutation of the purine degradation enzyme, **purine nucleoside phosphorylase**, results in the accumulation of the metabolite deoxy-GTP which is toxic to T-cell precursors through its ability to inhibit ribonucleotide reductase, an enzyme required for DNA synthesis and hence cell replication. Targeting of the T-cell lineage by this deficiency could well be linked to a relatively low level of 5′-nucleotidase. Some T-cells 'leak through' but they give inadequate protection against infection and the disease is usually fatal unless a bone marrow transplant life-line is offered.

Mutations in the recombinase enzymes, RAG-1 and RAG-2 (cf. figure 3.8), which initiate *VDJ* recombination events, usually result in complete failure to generate mature antigen-specific lymphocyte receptors and give rise to the severe combined immunodeficiency (SCID) phenotype (see below). In **Omenn's syndrome**, the particular RAG mutants allow some T-cells, apparently of Th2 phenotype, to sneak through, although they are not capable of preventing a failure to thrive and relatively early death.

MHC class II deficiency, previously known by its more sleazy appellation, 'bare lymphocyte syndrome', is associated with recurrent bronchopulmonary infections and chronic diarrhea occurring within the first year of life, with death from overwhelming viral infections at a mean age of 4 years. The condition arises from mutations affecting the RFX-B, CIITA, RFX-5 and RFXAP promoter proteins that bind to the 5′-untranslated region of the class II genes. Feeble expression of class II molecules on thymic epithelial cells grossly impedes the positive selection of CD4 T-helpers, and those that do leak through will not be encouraged by the lack of class II on antigen-presenting cells and B-lymphocytes. Note also that rare patients with mutations in *TAP-1* or *TAP-2* have an MHC class I bare lymphocyte syndrome.

Figure 15.6. A child with severe combined immunodeficiency showing skin lesions due to infection with vaccinia gangrenosum resulting from smallpox immunization. Lesions were widespread over the whole body. (Reproduced by kind permission of Professor R.J. Levinsky and the Medical Illustration Department of the Hospital for Sick Children, Great Ormond Street, London.)

Deficiencies leading to dysfunctional T-cells

Cell-mediated immunity (CMI) is depressed in immunodeficient patients with **ataxia telangiectasia** or with thrombocytopenia and eczema (**Wiskott–Aldrich syndrome**). The *W*iskott–*A*ldrich *s*yndrome *p*rotein (**WASP**) plays a critical role in signal transduction by interacting with Src-homology 3 (SH3) domains and in the regulation of cytoskeletal reorganization. WASP clusters physically with actin through the GTPase Cdc42 and possibly other molecules which regulate actin polymerization. Mutations in the *WASP* gene thus adversely affect cell motility, phagocyte chemotaxis, dendritic cell trafficking and the polarization of the T-cell cytoskeleton towards the B-cells during T–B collaboration. Poor cell-mediated immunity and impaired antibody production in affected boys are hardly surprising consequences.

Ataxia telangiectasia is a human autosomal recessive disorder of childhood which has been recognized as a **chromosomal breakage syndrome** characterized by progressive cerebellar ataxia with degeneration of Purkinje cells, a hypersensitivity to X-rays and an unduly high incidence of cancer. The mutated gene, *ATM*, encodes a protein kinase (Atm) which is a member of the phosphatidylinositol 3-kinase family. Following ultraviolet or ionizing radiation, the normal gene acts through the tumor suppressor protein p53 and thence p21 to arrest the cell cycle at the G1–S border, and through the Chk2 kinase and Cdc25 to block the G2–M transition. Presumably, the cellular DNA repair mechanisms now have a chance to operate. Regrettably, the relationship of the Atm defect to immunodeficiency is not clear. Another disease characterized by immune dysfunction, radiation sensitivity and increased incidence of cancer is the **Nijmegen breakage syndrome** where a mutation in the *NBS1* gene leads to a defective protein, nibrin, which normally functions as a component of a double-stranded DNA repair complex.

It is exciting to see the molecular basis of diseases being unraveled and an excellent example of nature yielding its secrets has been provided by studies on the **XL hyper-IgM syndrome**, a rare disorder characterized by recurrent bacterial infections, very low levels or absence of IgG, IgA and IgE and normal to raised concentrations of serum IgM and IgD. It transpires that point mutations and deletions in the T-cell CD40L (CD154) map to the part of the molecule thought to be concerned in the interaction with B-cell CD40 (cf. p. 174), thereby rendering the T-cells incapable of transmitting the signals needed for Ig class switching in B-cells. Maybe a note of caution regarding gene therapy would not be amiss at this point. CD40L knockout mice mimicking human X-linked hyper-IgM syndrome were injected with transduced BM or thymic cells using a retroviral vector containing the *CD40L* gene. Low level constitutive expression stimulated humoral and cellular immune functions, but 12 of 19 mice developed T-lymphoproliferative disorders, suggesting that current methods of gene therapy may be inappropriate for disorders involving highly regulated genes in essential positions in proliferative cascades.

Also for the collector are the rare cases of severe T-cell deficiency arising from mutation in the ε and γ chains of the CD3 complex and the ZAP-70 kinase (p. 166) which generate dysfunctional T-cells and preferentially impair the differentiation of CD8 T-cells.

COMBINED IMMUNODEFICIENCY

Severe *c*ombined *i*mmunodeficiency *d*isease (SCID) involving B-, T- and NK cells represents the most severe form of primary immunodeficiency, affecting one child in approximately every 80 000 live births. These infants exhibit profound defects in cellular and humoral immunity, with death occurring within the first year of life due to severe and recurrent opportunistic infections. Prolonged diarrhea resulting from gastrointestinal infections and pneumonia due to *Pneumocystis carinii* are common; *Candida albicans* grows vigorously in the mouth or on the skin. If vaccinated with attenuated organisms (figure 15.6), these infants usually die of progressive infection. SCID infants must be rescued by a bone marrow transplant if they are to survive.

Reticular dysgenesis is a rapidly fatal variant of severe combined immunodeficiency associated with a lack of both myeloid and lymphoid cell precursors, but SCID can arise from a variety of circumstances which selectively target lymphoid cell differentiation.

Mutations in the IL-7 signaling pathway cause SCID

Over half of the cases of severe combined immunodeficiency (SCID), which affects both B- and T-cell development, derive from mutations in the γ_c chain of the receptors for interleukins IL-2, -4, -7, -9 and -15. Of these, IL-7R is the most crucial for lymphocyte differentiation, and mutations in the interleukin-specific IL-7R α chain, or in JAK3 which transduces the γ_c signal, produce similar phenotypes. Mutations in the γ_c cytokine receptor gene were completely rescued with autologous bone marrow stem cells cultured with growth factors on fibronectin fragments (the magic ingredient needed for success) and transfected with a retroviral vector carrying the normal gene. While the T-cell defects can be corrected by transplantation of haplotype-matched allogeneic marrow within the first 3 months of life without the need for myeloablative pretreatment, and with survival rates of as high as 96%, donors may not always be readily accessible and, in a proportion of cases, the B- and NK cell defects are not corrected; these would benefit from the transfected autologous cell strategy.

SCID can arise from grossly deficient *VDJ* recombination

Unlike the sneak through of immunocompetent cells which accompanies the partial *RAG* deficiency in Omenn's syndrome, grossly dysfunctional mutations in the recombinase enzymes, which catalyse the introduction of the double-stranded breaks permitting subsequent recombination of the *V*, *D* and *J* segments, prevent the emergence of any mature lymphocytes. Failure of the *VDJ* recombination mechanism is also a feature of the radiosensitive cells from SCID patients with defective DNA-dependent protein kinase, a component of the complex which realigns and repairs the free coding ends created by the RAG enzymes. Just a snippet—naturally occurring SCID with a similar phenotype has been identified in Arabian foals.

SCID can be due to mutations in the purine salvage pathway enzyme, adenosine deaminase

Many SCID patients have a genetic deficiency of the purine degradation enzyme, adenosine deaminase (ADA), which results in the accumulation of the metabolite, dGTP, which is toxic to early lymphoid stem cells. The comparable immunodeficiency seen in acute lymphocytic leukemia patients treated with the ADA inhibitor deoxycoformacin attests to the validity of this analysis. Half the ADA-deficient SCID patients do reasonably well on transfusions of normal red cells containing the enzyme, whereas others with a longer standing more severe deficiency, which might have affected the thymus epithelium, also require treatment with the enzyme modified by polyethylene glycol which extends its half-life. These patients are excellent candidates for gene therapy. Children have been treated with periodic infusion of their own T-cells corrected by transfection with the *ADA* gene linked to a retroviral vector. Significant reconstitution of antibody responses and delayed-type skin tests to environmental antigens has been achieved without apparent complications. Hematopoietic stem cells from umbilical cord blood transfected with the *ADA* gene could be detected up to 18 months of age, but the level of expression was very low and, until the technology is improved, such patients must be maintained on enzyme replacement therapy.

Combined immunodeficiency resulting from inherited defective control of lymphocyte function

X-linked lymphoproliferative disease *(XLP)*, previously known as Duncan's syndrome, is a progressive immunodeficiency disorder characterized by fever, pharyngitis, lymphadenopathy and dysgammaglobulinemia, with a particular vulnerability to Epstein–Barr viral infection. Mutations have been uncovered in the *SH2DIA/SAP* gene encoding SAP (signaling lymphocytic activation molecule (SLAM)-*a*ssociated *p*rotein) which binds to SLAM through its SH2 domain. Since triggering of SLAM leads to strong induction of IFNγ in both Th1 and Th2 cells and acts on B-cells to enhance proliferation and increase susceptibility to apoptosis, mutations in SAP which adversely affect the activation of SLAM will weaken the immune response, especially with regard to EBV infection where viral replication with B-cells is heavily controlled by host T-cells. Mutation in another signal regulatory protein, the CD45 protein tyrosine phosphatase, gave rise to the SCID seen in a child with feeble lymphocyte responses at 2 months of age.

Mice with the *lpr* gene exhibit a **progressive lymphoproliferative syndrome** reflecting an accumulation of abnormal $CD4^-8^-$ T-cells with variable autoimmune features (cf. table 19.4). The mutation encodes an incompetent Fas (CD95) molecule and thereby defective Fas-triggered lymphocyte apoptosis. The homozygous deficiency is extremely rare in humans. Impaired T- and B-cell development and excessive production of autoantibodies by B-1 cells are the consequences of mutations in **motheaten** mice (cf. p. 417), affecting the *h*ematopoietic *c*ell *p*hosphatase (HCP) which regulates phosphorylation events triggered by engagement of antigen receptors.

RECOGNITION OF IMMUNODEFICIENCIES

Defects in immunoglobulins can be assessed by quantitative estimations; levels of 2 g/l arbitrarily define the practical lower limit of normal. The humoral immune response can be examined by first screening the serum for natural antibodies (A and B isohemagglutinins, heteroantibody to sheep red cells, bactericidins against *E. coli*) and then attempting to induce active immunization with diphtheria, tetanus, pertussis and killed poliomyelitis—but no live vaccines. CD19, 20 and 22 are the main markers used to enumerate B-cells by immunofluorescence.

Patients with T-cell deficiency will be hypo- or unreactive in skin tests to such antigens as tuberculin, *Candida*, trichophytin, streptokinase/streptodornase and mumps. Active skin sensitization with dinitrochlorobenzene may be undertaken. The reactivity of peripheral blood mononuclear cells to phytohemagglutinin is a good indicator of T-lymphocyte reactivity as is also the one-way mixed lymphocyte reaction (see Chapter 17). Enumeration of T-cells is most readily achieved by cytofluorimetry using CD3 monoclonal antibody.

In vitro tests for complement and for the bactericidal and other functions of polymorphs are available, while the reduction of nitroblue tetrazolium (NBT) or the stimulation of superoxide production provides a measure of the oxidative enzymes associated with active phagocytosis.

SECONDARY IMMUNODEFICIENCY

Immune responsiveness can be depressed nonspecifically by many factors. CMI in particular may be impaired in a state of malnutrition, even of the degree which may be encountered in urban areas of the more affluent regions of the world. Iron deficiency is particularly important in this respect, as are zinc and selenium deficiencies.

Viral infections are not infrequently immunosuppressive, and the profound fall in cell-mediated immunity which accompanies **measles infection** has been attributed to specific suppression of IL-12 production by viral cross-linking of monocyte surface CD46 (the complement regulator also known as membrane cofactor protein; cf. p. 307). The most notorious immunosuppressive virus, human immunodeficiency virus (HIV), will be elaborated upon in the next major section. In lepromatous leprosy and malarial infection there is evidence for a constraint on immune responsiveness imposed by distortion of the normal lymphoid traffic pathways and, additionally, in the latter instance, macrophage function appears to be aberrant. Skewing of the balance between Th1 and Th2 cells as a result of infection may also depress the subset most appropriate for immune protection.

Many therapeutic agents, such as X-rays, cytotoxic drugs and corticosteroids, although often used in a nonimmunological context, can nonetheless have dire effects on the immune system (see p. 382). **B-lymphoproliferative disorders**, such as chronic lymphatic leukemia, myeloma and Waldenström's macroglobulinemia, are associated with varying degrees of hypo-γ-globulinemia and impaired antibody responses. Their common infections with pyogenic bacteria contrast with the situation in Hodgkin's disease where the patients display all the hallmarks of defective CMI—susceptibility to tubercle bacillus, *Brucella*, *Cryptococcus* and herpes zoster virus.

ACQUIRED IMMUNODEFICIENCY SYNDROME (AIDS)

Clinical features

AIDS is a particularly unpleasant fatal disease caused by infection with the human immunodeficiency virus

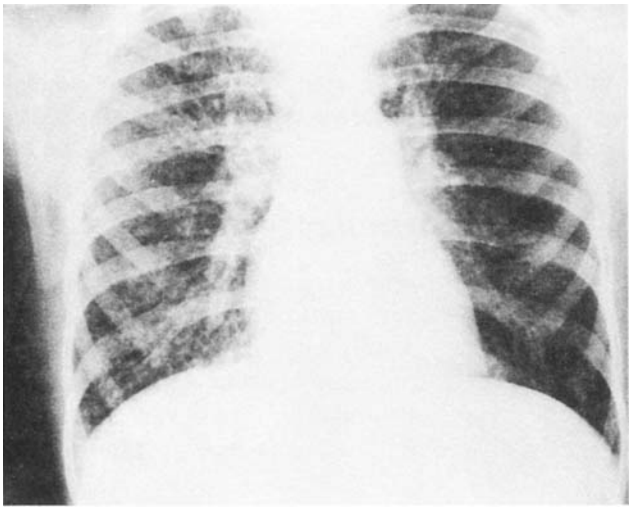


Figure 15.7. Viral pneumonia due to cytomegalovirus; radiograph showing extensive diffuse pneumonitis in both lung fields, characteristic of viral infection. (Reproduced from Lambert H.P. & Farrar W.E. (eds) (1982) *Infectious Diseases Illustrated*, p. 2.2. W.B. Saunders and Gower Medical Publishing, with permission.)

(HIV). It has reached endemic proportions and has thereby caused widespread alarm (and knowledge of immunology) amongst the public. Transmission is usually through infection with blood or semen containing HIV-1 or the related virus, HIV-2. In parts of Africa, where the incidence is frightful, transmission is largely by heterosexual contact. Prostitutes constitute a pivotal major initial reservoir of the infection, unlike their counterparts in developed countries, such as Japan, Australia, New Zealand, and the West, in whom the prevalence of AIDS has remained surprisingly low, and the majority of cases have occurred in male homosexuals, with other groups at risk including intravenous drug abusers, hemophiliacs receiving factor VIII derived from pooled plasma and infants of sexually promiscuous or drug-addicted mothers. Nevertheless, the number of infected heterosexuals is increasing. Death is usually due to pulmonary infection, but serious complications involving the nervous system are appearing in about 30% of cases. In essence, usually after a protracted latent period, there is a sudden onset of immunodeficiency associated with opportunistic infections involving, most commonly, *Pneumocystis carinii*, but also cytomegalovirus (figure 15.7), Epstein–Barr (EB) and herpes simplex viruses, fungi such as *Candida*, *Aspergillus* and *Cryptococcus*, and the protozoan *Toxoplasma*; additionally, there is exceptional susceptibility for Kaposi's sarcoma. There is also an AIDS-related complex (ARC), characterized by fever, weight loss and lymphadenopathy.

Characteristics of HIV

HIV-1/2 are members of the lentivirus group, which produce disease with a long latency and are adept at evading the immune system. They are RNA retroviruses with a relatively complex and tightly compressed genome (figure 15.8a) and the many virion proteins are generated by RNA splicing and cleavage by the viral protease (figure 15.8c).

The infection of cells by HIV

The envelope glycoprotein gp120 of HIV **binds avidly to cell surface CD4** molecules, but only initiates gp41-dependent fusion with the cell leading to infection when chemokine coreceptors are engaged (figure 15.8b). Helper T-cells with their abundant CD4 are a major target for infection, but the presence of even relatively low densities of CD4 on macrophages and microglia makes them susceptible to infection, and in the latter case is suspected to be a major factor in the cerebral complications of the disease. Different virus isolates vary in their relative tropism for T-cells and macrophages. Early in infection, the viruses utilize the CCR5 coreceptor for entry into memory T-cells and macrophages; these 'M-tropic' strains are usually unable to induce syncytia when cultured with CD4 T-cells (NSI, nonsyncytium inducing). These frequently evolve into 'T-tropic' strains which can infect resting T-cells using the CXCR4 coreceptor and are syncytium inducing (SI). Immature dendritic cells expressing CCR5 are attracted by the CC chemokines RANTES, MIP-1α and MIP-1β (see Table 10.3) which are released from cytotoxic T-lymphocytes (CTLs) when they encounter viral antigen. The dendritic cells capture HIV through a surface protein DC-SIGM and migrate to lymphoid tissue, where they form unique syncytia with T-cells which explosively drive replication of the virions by providing the two transcription factors needed: **Sp1** derived from unstimulated T-cells and **NFκB** proteins constitutively expressed by the dendritic cells. Undoubtedly, '*une liaison dangereuse*'! Possibly of crucial importance has been the discovery that suspensions of single *follicular* dendritic cells (FDCs) from human tonsil can be directly infected with HIV by a process which does not involve CD4. The infected cells permit viral replication and can reinfect T-cells *in vitro*. As we shall see, FDCs are a major reservoir of HIV in AIDS.

On gaining entry into a cell, HIV, as an RNA retrovirus, utilizes a **reverse transcriptase** to convert its genetic RNA into the corresponding DNA which is integrated into the host genome where it can remain

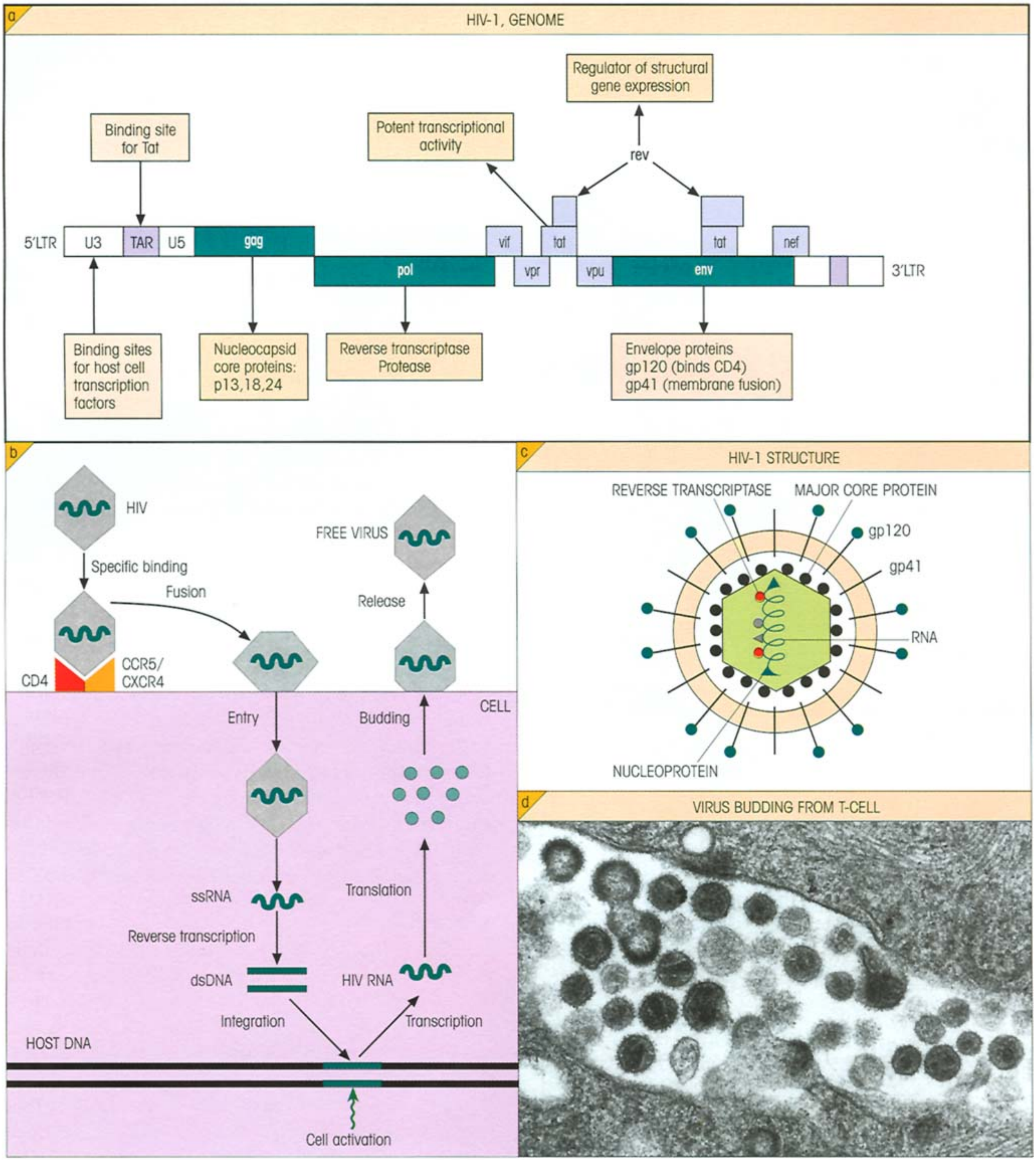


Figure 15.8. Characteristics of the HIV-1 AIDS virus. (a) HIV-1 genome and gene products. Tat, master switch, turns on all viral gene expression; *rev*, gene responsible for expression of structural proteins; *vif*, gene controlling infectivity by free, not cell-bound, virus; Vpr, weak transcriptional activator; Vpu, required for efficient virion budding and env processing; Nef, uncertain function linked to pathogenesis *in vivo*; LTR, long terminal repeats concerned in regulation of viral expression; contain regions reacting with products of *tat* and *nef* genes and in addition the *TATAAA* (the TATA box) promoter sequence, NFκB core enhancer elements and Sp-1-binding sites. Mice bearing the *tat* transgene develop Kaposi sarcoma-like skin lesions. (b) Intracellular life cycle of HIV. (c) HIV-1 structure. (d) Electron micrograph of mature and budding HIV-1 particles at the surface of human PHA blasts. (Courtesy of Dr Carol Upton and Professor S. Martin.)

latent for long periods (figure 15.8b). Stimulation of latently infected T-cells or macrophages activates HIV replication through an increase in the intracellular concentration of NFκB dimers, which bind to consensus sequences in the HIV enhancer region (figure 15.9). It is significant that tumor necrosis factor (TNF), which upregulates HIV replication through this NFκB pathway, is present in elevated concentrations in the plasma of HIV-infected individuals, particularly in the advanced stage. Perhaps, also, the greater susceptibility of Africans to AIDS may be linked to activation of the immune system through continual microbial insult. Infectious virus is finally released from the cell by budding (figure 15.8d).

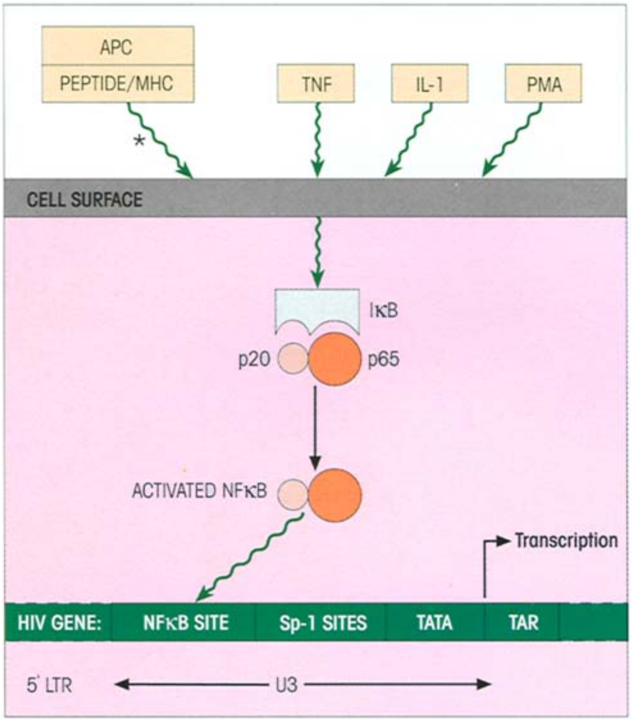


Figure 15.9. Upregulation of HIV replication in latently infected cell through external stimulation. *, T-cells only; IκB, inhibitor of NFκB.

Natural history of the disease

The sequence of events following HIV-1 infection is charted in figure 15.10. An acute early retroviral syndrome with fever, myalgia, arthralgia and so forth is accompanied by viremia and positive tests for blood p24, a dominant nucleocapsid antigen. An immune response to the virus identifiable by circulating antibodies to p24 and the envelope proteins gp120 and gp41, and by the production of gp120-specific cytotoxic T-cells, curtails the viremia and leads to **sequestration of HIV in lymphoid tissue**. Trapping of viral particles complexed with antibody and complement stimulates follicular hyperplasia and infection of the FDC. In effect, the follicles become the principal site for viral replication and infection of other cells of the immune system (figure 15.11). Eventually, follicular involution leads to a gradual degeneration of the FDC network with an increase in viral burden and replication in peripheral blood mononuclear cells. Crucially, in patients with progressive disease, **circulating CD4 T-cell numbers fall steadily** but, only when there is profound depletion with levels below, say, 50 mm^{-3}, do the consequences for CMI responses, as exemplified by the depressed delayed-type skin reactivity to common antigens (figure 15.12) and the failure to produce

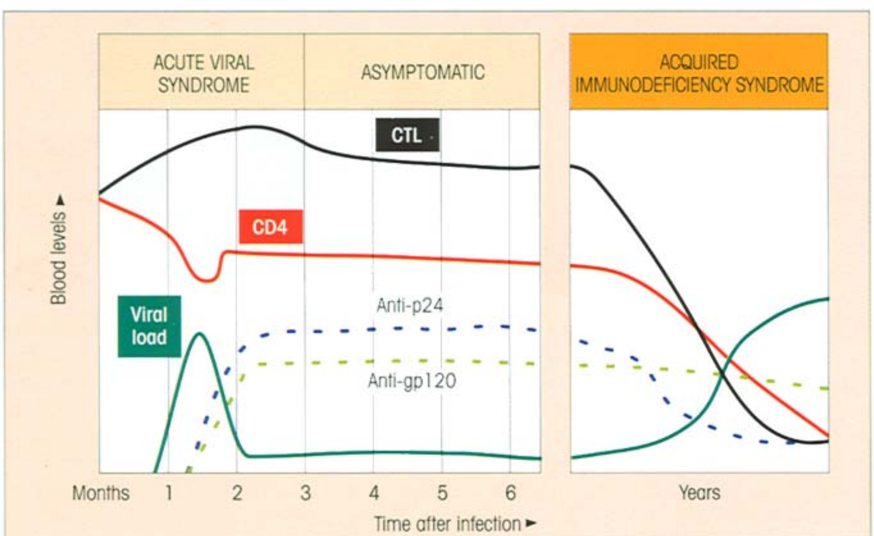


Figure 15.10. The natural history of HIV-1 infection.

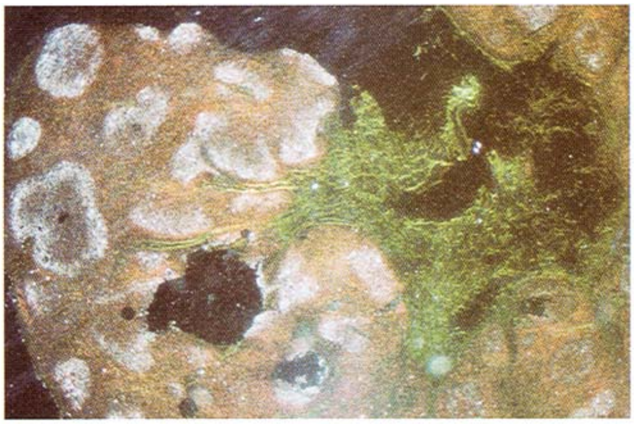
(a)   (b)

Figure 15.11. Abundance of HIV in lymphoid tissue demonstrated by *in situ* hybridization of lymph node sections from a representative HIV-infected patient in early stage disease. (a) Dark field image of a lymph node section after protease digestion. Location of HIV RNA is indicated by the silver grains which appear as white dots. An intense hybridization signal is predominantly restricted to the area of the germinal centers. (b) Higher magnification of a protease-digested section showing the intense distribution of silver grains in the light zone of a germinal center. (Reproduced from photos kindly provided by Dr A.S. Fauci with permission; cf. Pantaleo G., Graziosi C. & Fauci A.S. (1993) The role of the lymphoid organs in the pathogenesis of HIV infection. *Seminars in Immunology* **5**, 157.)

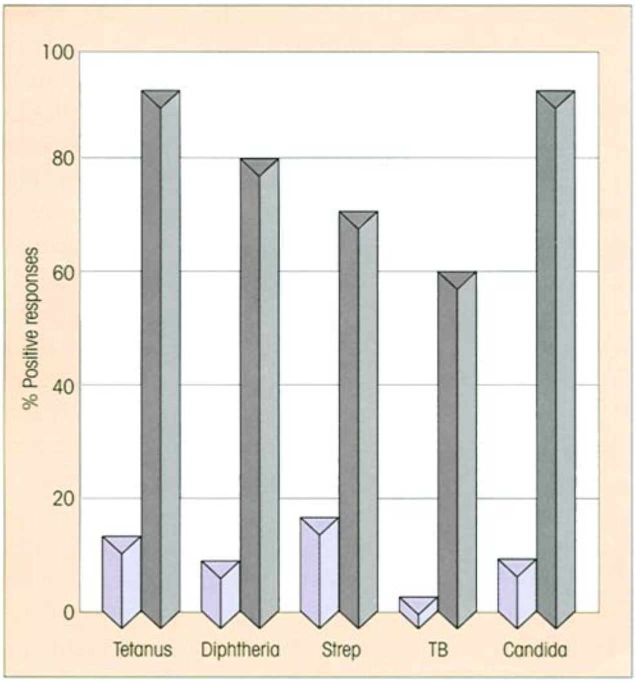


Figure 15.12. Skin test reactivity to common microbial antigens among AIDS patients (□; $n=20$) and controls (■; $n=10$). Other studies with more advanced patients have reported completely negative skin tests. (Reproduced from Lane H.C. & Fauci A.S. (1985) *Annual Reviews of Immunology* **3**, 477, with permission.)

cytomegalovirus-specific cytotoxic T-cells (Tc), become quite devastating. The patient is now wide open to life-threatening infections caused by normally non-pathogenic (i.e. opportunistic) agents such as *Pneumocystis carinii* and cytomegalovirus, characteristic of AIDS.

The diagnostic features of AIDS are pretty unmistakable

An individual with opportunistic infections, lymphopenia, low CD4 but relatively normal CD8 in the peripheral blood, raised IgG and IgA levels and poor skin tests to common recall antigens (figure 15.12) may well have AIDS, particularly if he or she comes from a group at risk. γ-Interferon (IFNγ) and neopterin, a degradation product of guanosine triphosphate (GTP) induced in macrophages by IFNγ, are good indicators of CMI and are significantly increased in AIDS infection preceding a subsequent loss of CD4 cells. Confirmation of the diagnosis comes from lymph node biopsy, showing profound abnormalities and drastic changes in germinal centers, and the demonstration of viral antibodies by enzyme-linked immunosorbent assay (ELISA) and by 'immunoblotting' (cf. pp. 113 and 117). Serum p24 antigen is often positive in active disease but isolation of virus or demonstration of the HIV genome provides the final confirmation of infection.

Factors affecting the progression of AIDS (figure 15.13)

By looking for correlations between the speed with which infected individuals develop disease and the various parameters of the immune response, one hopes to gain insight into those elements which afford

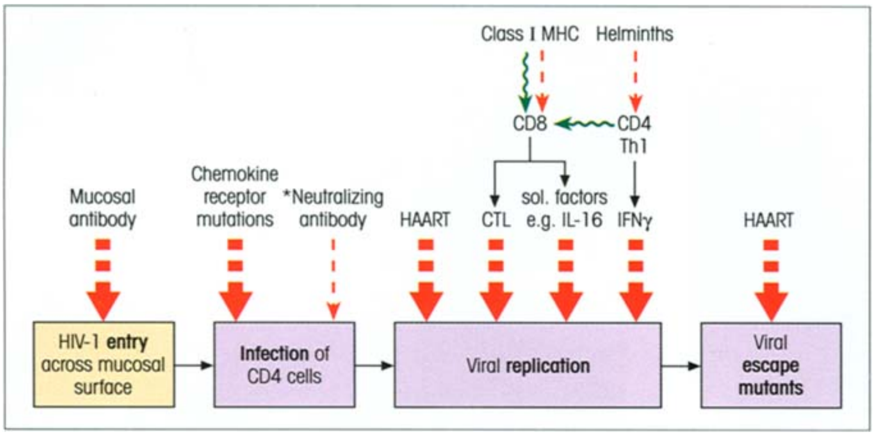


Figure 15.13. Factors affecting HIV progression. *Neutralizing antibody more effective against HIV-2. HAART, highly active antiretroviral drug therapy.

some degree of protection and could guide the attempts to produce effective vaccines. Bearing in mind the discovery of the importance of CCR5 as a chemokine coreceptor for the infection of cells with NSI HIV clades, it is gratifying to note the 10-fold reduced risk of HIV infection in individuals **homozygous for the *CCR5Δ32* allele**, where the 32 base deletion encodes a nonfunctional product. Homozygotes represent 1–3% of northern Europeans, declining as one goes south, and absent in African, Asiatic and Oceanic populations. It has been speculated that the mutation may have been fixed in northwest Europe in the 14th century to provide a selective advantage against bubonic plague. The progression of HIV infection is delayed even in the heterozygotes.

As mentioned above, a powerful early **CD8 CTL response** can drastically reduce the viral load with beneficial effects long term. An important role for CTLs is implied by the extended survival of 28–40% of HIV-1-infected Caucasians who were fully heterozygous at HLA class I loci *lacking* B35 or CW4 alleles or both. Furthermore, viral load increased dramatically following CD8 depletion in macaque monkeys infected with simian immunodeficiency virus (SIV) which causes an AIDS-like syndrome. Being an RNA retrovirus, HIV is subject to a ferociously high mutation rate, and longitudinal studies in this model showed that CTL evasion by epitope mutation is an important escape strategy for compromising the antiviral immune response.

In addition to their capacity as CTL effectors, CD8 cells from infected patients produce a number of factors which suppress HIV replication in cultured CD4 cells. One of these factors is IL-16 which cross-links the membrane-proximal D4 domain of CD4 and blocks infection by the virus. It is relevant that high serum levels of IL-16 correlate well with slow progression of the disease.

Enumeration with HLA class I/peptide tetramers has revealed that up to 5% of total CD8 cells in the acute phase are specific for a single epitope and the numbers inversely correlate with viral load. As time goes on, however, the protective power of the cells is not commensurate with their numbers; in boxing jargon, they 'punch below their weight'. Whether they are purely dysfunctional or represent anergic or regulatory IL-10-producing cells is not clear, but on one point there is a general consensus: a sustained healthy **CD8 response depends heavily on robust CD4 Th1 effectors** contributing IFNγ, chemokines and useful cell to cell interactions. And here is the root of the problem since, initially, although more than 0.5% of total CD4 cells are specific for viral proteins, these are deleted soon after infection. There is no shortage of hypotheses to account for this depletion, but money tends to be riding on the induction of apoptosis in proliferating antigen-specific CD4 cells infected either directly or through the intimacy of syncytium formation with dendritic cells harboring the virus.

The horrendous fall in overall numbers of CD4 cells associated with the onset of clinical AIDS may be precipitated by the evolution of T-cell syncytium inducing (SI) strains as the disease progresses. The ability of SI virus to infect thymocytes might exacerbate an already precarious situation by cutting off the admittedly low volume supply of new T-cells still generated by an aged thymus. (Enthusiastic readers can ferret out even more hypothetical mechanisms for HIV-mediated T-depletion, e.g. in *Essential Immunology*, 9th edn, pp. 323–324, and Hazenberg M.O. *et al.* (2000) *Nature Immunology* **1**, 285.)

Antibodies to variable loop 3 (V3) of the envelope gp120 protein arise during disease and are frequently capable of preventing viral infection of CD4 T-cells *in vitro*. Nonetheless, they are rarely protective *in* *vivo*, partly because monkey experiments showed that very high titers are required and masking of epitopes by overhanging carbohydrate groups may blunt the immunogenicity, perhaps because virus may be directly transmitted from follicular dendritic cells to CD4 T-cells without exposure to antibody, and certainly because the V3 sequences undergo extensive mutation.

Neutralizing antibodies are more effective in **HIV-2** infection where progression to death is much slower than with HIV-1, and asymptomatic individuals commonly reveal deletions in the *Nef* gene, a slow or negligible decline in CD4 cells and strong CTL responses.

Therapeutic strategies for controlling AIDS

Highly active antiretroviral drug therapy (HAART) is very effective when given early in the disease and can reduce plasma viral load to undetectable levels. But the virus persists in a latent form in resting CD4 T-cells with a decay half-life of 40 months. Supposing that the reservoir of latent cells is as small as 1×10^5 cells, effective eradication could take as long as 60 years and, in any case, the treatment is very costly and not without side-effects. The sensible strategy is to keep this residual virus under control by the immune system. One approach is to use an intermittent drug regimen. This allows a re-emergence of virus capable of boosting reasonably robust immune responses which, in turn, would synergize with the reinstallation of combination drug therapy. The goal, though, is to develop a vaccine which could contain the infection following the initial HAART without the need for further antiviral drugs.

Given the relative efficacy of CD8 T-cells early in infection and their dependence on robust CD4 helpers, together with the evidence for the protective role of cell-mediated immunity in the SIV rhesus monkey model, effort is being directed towards Th1-biased vaccines. Logically, this should go hand-in-hand with the eradication of any concurrent worm infestation which would favor Th2 responses. Trials are in progress with vaccines incorporating a large number of T-cell epitope peptide sequences to minimize the chance of the infecting strains evading immune recognition by mutation and, where prophylactic immunization is contemplated, to take advantage of the fair degree of cross-clade recognition of defined CTL epitopes. In a series of studies in rhesus macaques, it transpired that the most promising containment of challenge by a highly virulent virus with an envelope heterologous to the immunizing strain was achieved by intradermal priming with envelope *gag*, *pol* and *Nef* DNA, followed by boosting with recombinant fowlpox virus. Oral immunization with an **attenuated** recombined SIV/HIV vaccine in which the *Vpu* and *Nef* genes were deleted led to protection against vaginally transmitted infection, but there was a disturbing tendency for even the attenuated strains to be pathogenic. This underscores the importance of mucosal defense against initial infection and must give some credence to the idea that the presence of antiviral antibodies in mucosal secretions could be a significant factor in primary defense. In fact, despite the discouraging comments raised earlier, the identification of a strongly neutralizing monoclonal antibody directed to a conserved region of the gp160 envelope precursor suggests that, in a normal infection, antibodies to this potentially protective epitope might be overwhelmed by non-neutralizing responses to other adjacent B-cell epitopes; if this were so, an epitope which mimicked this conserved epitope could be an additional valuable vaccine candidate.

Some other approaches, still in their formative stages, are in the wings. Now that techniques of transfecting bone marrow cells are more successful, one looks for advances in 'intracellular immunization' through the introduction of genes encoding potentially anti-HIV molecules which will protect the future differentiated CD4 T-cell from infection. In one approach, bone marrow cells are transfected with a sequence of the CD4 gene coding for an HIV-blocking peptide; on regrafting, some of the stem cells will become $CD4^+$ T-cells, secreting a surrounding barrier of the blocking peptide which protects the lymphocyte from infection by the virus. Another group have transfected the stem cells with an scFv (cf. p. 125) gene encoding a single HIV-specific antibody combining site which disrupts the viral replication cycle. Using a similar system, one could also envisage the transfection of antisense genes to block the expression of HIV regulatory genes. Despite this frantic activity, a change in social behavior patterns would have a major impact on the relentless spread of AIDS. One cannot help recalling the aggrieved request by God to the supplicant, frustrated by the failure of his prayers to win him a lottery prize, to help Him by buying a lottery ticket.

SUMMARY

Primary immunodeficiency states (table 15.1)

- These occur in the human, albeit somewhat rarely, as a result of a defect in almost any stage of differentiation in the whole immune system.
- Rare X-linked mutations produce disease in males.
- Defects in phagocytic cells, the complement pathways or the B-cell system lead in particular to infection with bacteria which are disposed of by opsonization and phagocytosis.
- Patients with T-cell deficiencies are susceptible to viruses and moulds which are normally eradicated by CMI.

Table 15.1. Summary of primary immunodeficiency states.

DEFECTIVE GENE PRODUCT(S)	DISORDER
COMPLEMENT DEFICIENCES	
C1, C2, C4	Immune complex disease (SLE)
C1 inhibitor	Angioedema
DAF, HRF, CD59 (GPI anchor regulatory proteins)	Paroxysmal nocturnal hemoglobinuria
C3, Fact H, Fact I	Recurrent pyogenic infections
C5, C6, C7, C8 Properdin	Recurrent *Neisseria* infections
PHAGOCYTIC DEFECTS	
NADPH oxidase	Chronic granulomatous disease
CD18 (β_2-integrin β chain)	Leukocyte adhesion deficiency
?	Chediak–Higashi disease
IFNγR1/2, IL-12 p40, IL-12Rβ1	Mendelian susceptibility to mycobacterial infection
TNFR	TNF-receptor associated periodic syndrome
Pyrin	Familial Mediterranean fever
PRIMARY B-CELL DEFICIENCY	
Bruton's tyrosine kinase	Bruton congenital a-γ-globulinemia
μ heavy chain, λ_5 chain	Pro-B arrest; Bruton phenotype
?	IgA deficiency and common variable immunodeficiency
?	Transient infant hypo-γ-globulinemia
PRIMARY T-CELL DEFICIENCY	
?	DiGeorge and Nezelof syndromes; failure of thymic development
PNP	Purine nucleoside phosphorylase deficiency toxic to T-cells
RAG-1/2	Omenn's syndrome; partial *VDJ* recombination
MHC class II promoters	'Bare lymphocyte syndrome'
WASP	Wiskott–Aldridge syndrome; defective cytoskeletal regulation
Atm	Ataxia telangiectasia; defective DNA repair
CD154 (CD40L)	X-linked hyper-IgM syndrome
CD3 ε and γ chains, ZAP-70	Severe T-cell deficiency
COMBINED IMMUNODEFICIENCY	
?	Reticular dysgenesis; defective production of myeloid and lymphoid precursors
IL-7Rα, γ_c, Jak3	SCID due to defective IL-7 signaling
RAG-1/2	Complete failure of *VDJ* recombination
ADA	Adenosine deaminase deficiency; toxic to early lymphoid stem cells
SAP	X-linked lymphoproliferative disease; defective cell signaling

(*continued*)

Secondary immunodeficiency

- Immunodeficiency may arise as a secondary consequence of malnutrition, lymphoproliferative disorders, agents such as X-rays and cytotoxic drugs, and viral infections.

Acquired immunodeficiency syndrome (AIDS)

- AIDS results from infection by the RNA retroviruses HIV-1 and HIV-2.
- HIV infects T-helper cells through binding of its envelope gp120 to CD4 and either CCR5 or CXCR4 chemokine receptor cofactors. It also infects macrophages, microglia, T-cell-stimulating dendritic cells and FDCs, the latter through a CD4-independent pathway.
- Within the cell, the RNA is converted by the reverse transcriptase to DNA, which can be incorporated into the host's genome, where it lies dormant until the cell is activated by stimulators such as TNF which increase NFκB levels.
- There is usually a long asymptomatic phase after the early acute viral infection has been curtailed by a CD8 CTL immune response and the virus is sequestered to the FDC in the lymphoid follicles where it progressively destroys the dendritic cell meshwork.
- A disastrous fall in CD4 T-helpers destroys cell-mediated defenses and leaves the patient open to life-threatening infections through opportunist organisms such as *Pneumocystis carinii* and cytomegalovirus.
- There is a tremendous battle between the immune system and the virus, with extremely high rates of viral destruction and CD4 T-cell replacement.
- CD4 T-cell depletion may eventually occur as a result of direct pathogenicity, syncytium formation, susceptibility to apoptosis and possibly other mechanisms.
- AIDS is diagnosed in an individual with opportunistic infections, by low CD4 but normal CD8 T-cells in blood, poor delayed-type skin tests, positive tests for viral antibodies and p24 antigen, lymph node biopsy and isolation of live virus or demonstration of HIV genome by the polymerase chain reaction (PCR).
- Highly active antiretroviral drug therapy (HAART), combining inhibitors of reverse transcriptase and protease, can eliminate detectable virus early in disease, although latent virus remains.
- Vaccines are being targeted to Th1 responses but it is a very difficult virus to control.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

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Hypersensitivity

INTRODUCTION

When an individual has been immunologically primed, further contact with antigen leads to secondary boosting of the immune response. However, the reaction may be excessive and lead to gross tissue changes (hypersensitivity) if the antigen is present in relatively large amounts or if the humoral and cellular immune state is at a heightened level. It should be emphasized that the mechanisms underlying **these inappropriate immune responses leading to tissue damage**, which we speak of as **hypersensitivity reactions**, are those normally employed by the body in combating infection as discussed in Chapter 13. Hypersensitivity can also arise from direct interaction of the inciting agent with elements of the innate immune system without intervention by acquired responses.

ANAPHYLACTIC HYPERSENSITIVITY (TYPE I)

The phenomenon of anaphylaxis

The earliest accounts of inappropriate responses to foreign antigens relate to **anaphylaxis** (Milestone 16.1). The phenomenon can be readily reproduced in guinea-pigs which, like humans, are a highly susceptible species. A single injection of 1 mg of an antigen such as egg albumin into a guinea-pig has no obvious effect. Repeat the injection 2–3 weeks later and the sensitized animal reacts very dramatically with the symptoms of generalized anaphylaxis; almost immediately, the guinea-pig begins to wheeze and within a few minutes dies from asphyxia. Examination shows intense constriction of the bronchioles and bronchi and generally there is: (i) contraction of smooth muscle, and (ii) dilatation of capillaries. Similar reactions do occur in human subjects and have been observed following wasp and bee stings or injections of penicillin in appropriately sensitive individuals. In many instances only a timely injection of epinephrine to counter the smooth muscle contraction and capillary dilatation can prevent death.

Sir Henry Dale recognized that histamine mimics the systemic changes of anaphylaxis and, furthermore, that exposure of the uterus from a sensitized guinea-pig to antigen induces an immediate contraction associated with an explosive degranulation of mast cells (figure 1.14) responsible for the release of histamine and a number of other mediators (figure 1.15).

Anaphylaxis is triggered by clustering of IgE receptors on mast cells through cross-linking

Two main types of mast cell have been recognized, exemplified in the rat by those in the intestinal mucosa and those in the peritoneum and other connective tissue sites. They differ in a number of respects, for example in the type of protease and proteoglycan in their granules, and in the proliferative response of the mucosal mast cell to the T-cell cytokine IL-3 (table 16.1). This last point is made rather tellingly by the striking proliferation of mast cells in the intestinal mucosa dur-

Milestone 16.1—The Discovery of Anaphylaxis

People are not equal. Idiosyncratic responses to given stimuli have been recognized from time immemorial and hypersensitive reactions in some individuals to normally innocuous environmental agents have been frequently observed. As Lucretius remarked two thousand years ago, 'Differences are so great that one man's meat is another man's poison'. Sir Thomas More records that the future King Richard III was aware that strawberries gave him urticaria and arranged to be served a bowl of the fruit at a banquet attended by an arch-enemy. When he broke out in a spectacular rash, he accused his guest of attempted poisoning and had him summarily executed.

Scientific interest in hypersensitivity was aroused by the observations of Richet and Portier. During a South Sea cruise on Prince Albert of Monaco's yacht, the Prince, presumably smarting from an encounter with *Physalia* (the jelly-fish known as the Portugese-Man-of-War with very nasty tentacles), suggested that toxin production by the fish might be of interest. Let Richet and Portier take up the story in their own words (1902).

'On board the Prince's yacht, experiments were carried out proving that an aqueous glycerin extract of the filaments of *Physalia* is extremely toxic to ducks and rabbits. On returning to France, I could not obtain *Physalia* and decided to study comparatively the tentacles of *Actinaria* (sea anemone). While endeavouring to determine the toxic dose (of extracts), we soon discovered that some days must elapse before fixing it; for several dogs did not die until the fourth or fifth day after administration or even later. We kept those that had been given insufficient to kill, in order to carry out a second investigation upon these when they had recovered. At this point an unforeseen event occurred. The dogs which had recovered were intensely sensitive and died a few minutes after the administration of small doses. The most typical experiment, that in which the result was indisputable, was carried out on a particularly healthy dog. It was given at first 0.1 ml of the glycerin extract without becoming ill: 22 days later, as it was in perfect health, I gave it a second injection of the same amount. In a few seconds it was extremely ill; breathing became distressful; it could scarcely drag itself along, lay on its side, was seized with diarrhea, vomited blood and died in 25 minutes.'

The development of sensitivity to relatively harmless substances was termed by these authors **anaphylaxis**, in contrast to **prophylaxis**.

Table 16.1. Comparison of two types of mast cell.

CHARACTERISTICS	MUCOSAL MAST CELL	CONNECTIVE TISSUE MAST CELL
GENERAL		
Abbreviation*	MC_t	MC_{tc}
Distribution	Gut & lung	Most tissues**
Differentiation favored by	IL-3	Fibroblast factor
T-cell dependence	+	–
High affinity Fcε receptor	2×10^5/cell	3×10^4/cell
GRANULES		
Alcian blue and Safranin staining	Blue & brown	Blue
Ultrastructure	Scrolls	Gratings/lattices
Protease	Tryptase	Tryptase & chymase
Proteoglycan	Chondroitin sulfate	Heparin
DEGRANULATION		
Histamine release	+	++
LTC_4 : PGD_2 release	25 : 1	1 : 40
Blocked by disodium cromoglycate/theophylline	–	+

*Based on protease in granules.
**Predominate in normal skin and intestinal submucosa.

ing infection with certain parasites in intact, but not in T-depleted, rodents, the effect being mediated by a combination of IL-3 and IL-4. The two types have common precursors and are interconvertible depending upon the environmental conditions, with mucosal MC_t (*t*ryptase) phenotype favored by IL-3 and connective tissue MC_{tc} (*t*ryptase *c*hymotryptase) being promoted by a fibroblast factor. However, both types display a high affinity receptor for IgE (FcεRI: cf. figure 3.18), a property shared with their circulating counterpart, the basophil. The strength of binding to the mast cell is evident from the retention of IgE antibodies at a site of intradermal injection for several weeks; IgG4 in the human also binds to the mast cell receptor, but more weakly, and disperses from the injection site within a day or so. It has long been established that the anaphylactic antibodies in the human are mainly of the IgE class.

Anaphylaxis is mediated by the reaction of allergen with IgE antibodies bound strongly through their Cε3 domains to the α chain of the high affinity receptor (FcεRI; $K_d = 10^{-9}$ to 10^{-10} M) on the surface of the mast cell (figure 16.1). Cross-linking of these IgE antibodies by a multivalent hapten will trigger media-

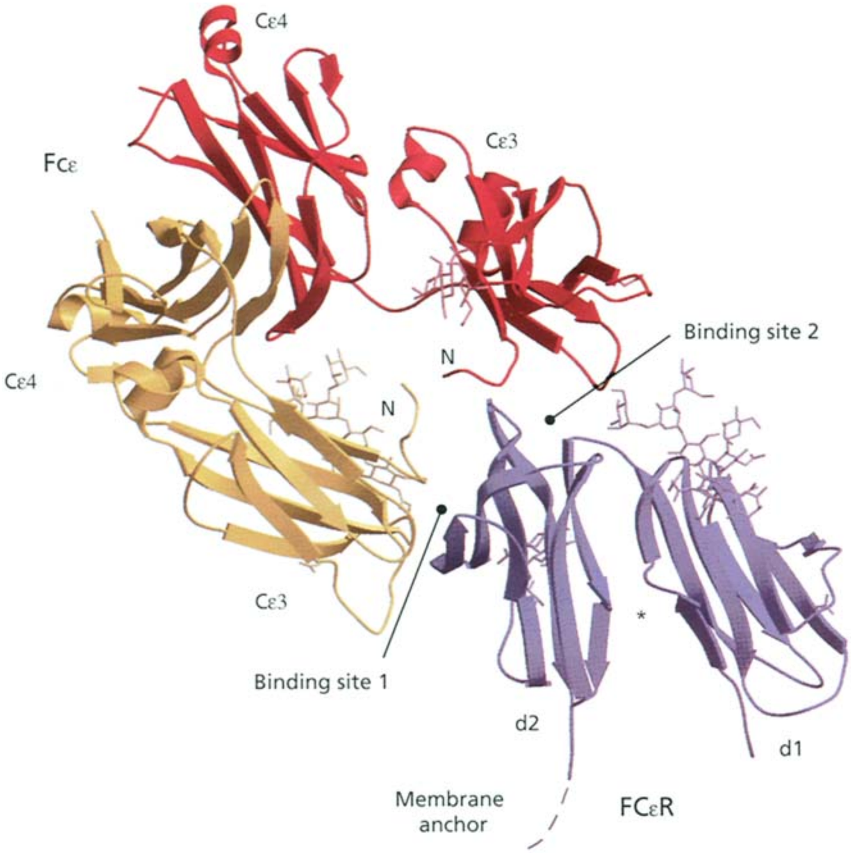


Figure 16.1. The structural basis of the binding of IgE to the high affinity mast cell receptor FcεRI. Side view of the complex with the two Fc chains in yellow and red and the FcεRIα chain in blue; carbohydrate residues are shown as sticks. The two Cε3 domains of the heavy chain dimer of IgE bind asymmetrically to two distinct interaction sites on the α chain of the receptor. The β-turn loop on one Cε3 binds along one side of the d2 domain, while surface loops plus the Cε2–Cε3 linker region on the other Cε3 interact with the top of the d1–d2 interface. The 1 : 1 stoichiometry of this asymmetric binding precludes the linkage of one IgE to two receptor molecules and ensures that triggering due to α–α aggregation only occurs through multivalent binding to surface IgE (see figure 16.2). The cleft between the d2 and d1 domains facing the plasma membrane is probably connected in some way to the receptor β chain to provide information on the aggregation status of the α chain. (Photograph kindly provided by Dr Ted Jardetzky and reproduced by permission of the Nature Publishing Group.)

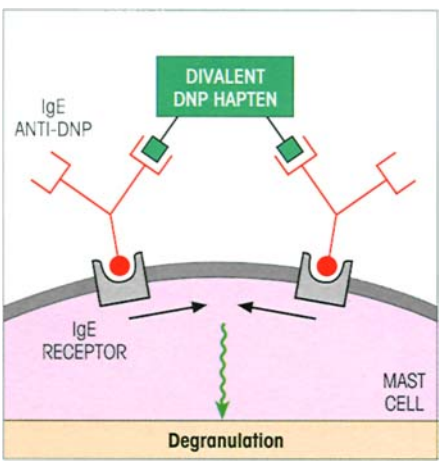


Figure 16.2. Clustering of IgE receptors either by multivalent hapten or antibody to the receptors themselves leads to mast cell degranulation. Studies using preformed aggregates of IgE of known size which stimulate mast cells showed that the reaction could be terminated by addition of selective kinase inhibitors. Small stable clusters of receptors continue to initiate signals for at least 1 hour. Lyn, which is the first tyrosine kinase to be activated, is normally present in caveolae membrane domains and only cosediments with the FcεRI when the latter are aggregated.

tor release (figure 16.2); trimers are more effective than dimers and tetramers even more so. The critical event is aggregation of the receptors by cross-linking as clearly shown by the ability of divalent antibodies reacting directly with the receptor to trigger the mast cell.

The FcεRIγ subunit resembles the ζ chain of the T-cell receptor (TCR) and part of the low affinity FcγRIII on macrophages. The cytoplasmic domain on the β and γ subunits shares a common *i*mmunoreceptor *t*yrosine-based *a*ctivation *m*otif (ITAM) with CD3γ, δ, ε, TCR ζ, η and B-cell receptor-associated Ig-α and Ig-β, and truncation of this domain abolishes receptor-mediated activation in a reconstituted system. Aggregation of the FcεRIα chains through cross-linking of the bound IgE antibodies activates the Lyn protein tyrosine kinase associated with the β chains and, if the aggregates persist, this leads to transphosphorylation of the β and γ chains of other receptors within the cluster and recruitment of the Syk kinase (figure 16.3). The subsequent series of phosphorylation-induced activations leads to the recruitment of the phospholipase C and the GTPase-linked Ras and Rac pathways as set out in figure 16.3. The net result of the biochemical cascade is degranulation with release of preformed mediators and the synthesis of arachidonic acid metabolites

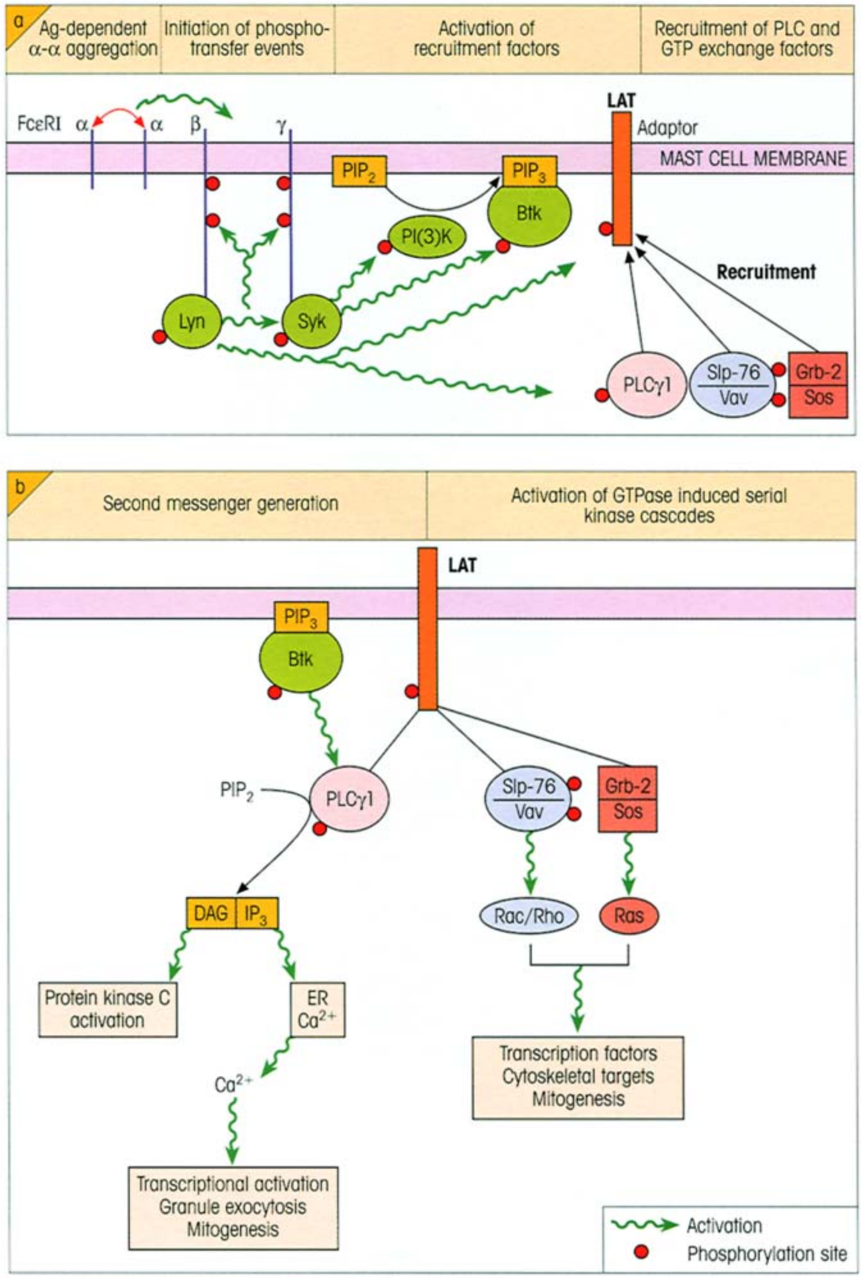


Figure 16.3. Mast cell triggering. (a) Early events in signaling through the high affinity IgE receptor, FcεRI. Aggregation of the FcεRIα chains through cross-linking of bound IgE by multivalent antigen (allergen) alerts the β and γ chains of the receptor, leading to activation of the linked Lyn and Syk protein tyrosine kinases. They in turn phosphorylate and activate the PI(3) kinase, Bruton's tyrosine kinase (Btk) and the membrane adaptor protein LAT which recruits phospholipase Cγ1 and adaptor molecules concerned in the activation of GTPase/kinase cascades. (b) The activated phospholipase Cγ1 generates diacylglycerol (DAG) which targets protein kinase C, while inositol (1,4,5)P$_3$ (IP$_3$) elevates cytoplasmic Ca^{2+} by depleting the ER stores. The raised calcium concentration activates transcriptional factors and causes granule exocytosis. The Grb-2/Sos and Slp-76/Vav complexes also associated with the LAT adaptor trigger the Ras and Rac/Rho GTPase-induced serial kinase cascades, respectively, leading to the activation of transcription factors and rearrangements of the actin cytoskeleton. (Figure essentially designed by Dr Helen Turner, based on the article by Turner H. & Kinet J.-P. (1999) *Nature* (Supplement on Allergy and Asthma) **402**, B24.)

formed by the cyclo-oxygenase and lipoxygenase pathways (cf. figure 1.15). To recapitulate, the preformed mediators released from the granules include histamine, heparin, neutral protease, eosinophil and neutrophil chemotactic factors and platelet activating factor, while leukotrienes LTB4, LTC4 and LTD4, the prostaglandin PGD2 and thromboxanes are all newly synthesized. We now know that IL-3, IL-4, IL-5 and IL-6 and *g*ranulocyte–*m*acrophage *c*olony-*s*timulating *f*actor (GM-CSF), a typical Th2 pattern of cytokines cells, are also released.

Under normal circumstances, these mediators help to orchestrate the development of a defensive acute inflammatory reaction (and in this context let us not forget that complement fragments C3a and C5a can also trigger mast cells, although not through IgE receptors). When there is a massive release of these mediators under abnormal conditions, as in atopic disease, their bronchoconstrictive and vasodilatory effects predominate and become distinctly threatening.

Atopic allergy

Clinical responses to extrinsic allergens

At least 10% of the population suffer to a greater or lesser degree with allergies involving localized IgE-mediated anaphylactic reactions to allergens such

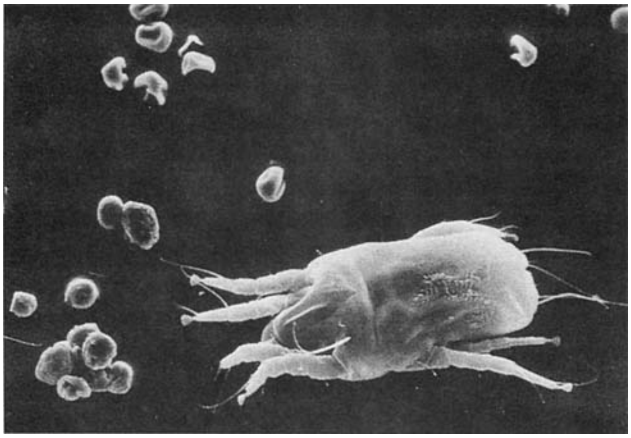


Figure 16.4. House dust mite—a major cause of allergic disease. The electron micrograph shows the rather nasty looking mite graced by the name *Dermatophagoides pteryonyssinus* and fecal pellets on the bottom left which are the major source of allergen. The biconcave pollen grains (top left) shown for comparison indicate the size of particle which can become airborne and reach the lungs. The mite itself is much too large for that. (Reproduced by courtesy of Dr E. Tovey.)

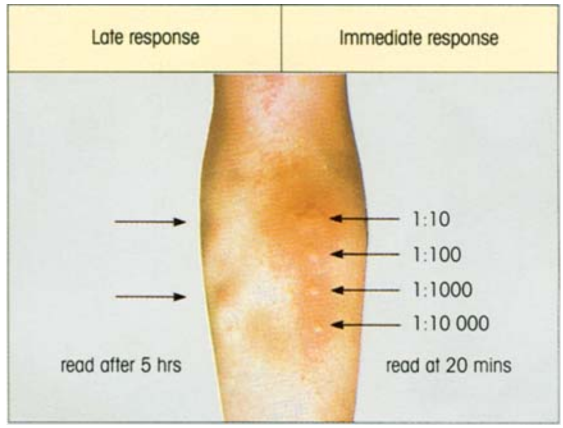


Figure 16.5. Atopic allergy. Skin prick tests with grass pollen allergen in a patient with typical summer hay fever. Skin tests were performed 5 hours (left) and 20 minutes (right) before the photograph was taken. The tests on the right show a typical end-point titration of a type I immediate wheal and flare reaction. The late phase skin reaction (left) can be clearly seen at 5 hours, especially where a large immediate response has preceded it. Figures for allergen dilution are given.

as grass pollens, animal danders, the feces from mites in house dust (figure 16.4) and so on. An increasing number of allergens have now been cloned and expressed including **Der p1** from mites and **Lol pI–V** from rye grass pollen. Der p1 proves to be a protease which increases the permeability of the bronchial mucosa, thereby facilitating its own passage along with other allergens across the epithelium and allowing access to and sensitization of cells of the immune system. It splits the low affinity IgE receptor, CD23, on B-cells so reducing its negative impact on IgE synthesis when occupied by IgE; it also cleaves the IL-2 receptor α chain which biases the immune response to Th2-dependent IgE production. Short cuts to allergen purification can be achieved by screening cDNA libraries for IgE-binding proteins using immunoblotting techniques. This was a godsend for the purification of the allergen from the venom of the Australian jumper ant, *Myrmecia pilosula*; just think of trying to accumulate ants by the kilogram to isolate the allergen using conventional protein fractionation.

The local anaphylactic reaction to injection of antigen into the skin of atopic patients is manifest as a wheal and flare (figure 16.5) which is short lived and resolves within an hour or so; it may be succeeded by a late phase response involving eosinophil infiltrates which peak at around 5 hours. Contact of the allergen with cell-bound IgE in the bronchial tree, the nasal mucosa and the conjunctival tissues releases mediators of anaphylaxis and produces the symptoms of **asthma** or **allergic rhinitis and conjunctivitis** (hay fever) as the case may be. A proportion of the patients who experience late phase responses after bronchial challenge with allergen eventually develop chronic **asthma**. This disease affects 155 million individuals worldwide and costs $6 billion a year to treat in the USA alone. Patients fall into three main categories.

1 The majority who are **extrinsic asthmatics** associated with **atopy**, i.e. the genetic predisposition to synthesize inappropriate levels of IgE specific for external allergens.

2 Nonatopic intrinsic asthmatics.

3 **Occupational asthmatics** exposed to specific proteins or small molecular weight chemicals.

Bronchial biopsy and lavage of asthmatic patients reveal an unequivocal involvement of **mast cells and eosinophils** as the major mediators secreting effector cells, while T-cells provide the microenvironment required to sustain the chronic inflammatory response which is an essential feature of the histopathology in each category (figure 16.6). The resulting variable airflow obstruction and bronchial hyper-responsiveness are the cardinal clinical and physiological features of the disease.

The atopic trait can also manifest itself as an **atopic dermatitis** (figure 16.7), with house dust mite, domestic cats and German cockroaches often proving to be the environmental offenders. Recalling the inflammation in asthma, skin patch tests with Der p1 in these eczema patients produce an infiltrate of eosinophils, T-cells, mast cells and basophils. The number of individuals affected is comparable to the number affected by asthma which may be a surprise to some.

Awareness of the importance of IgE sensitization to

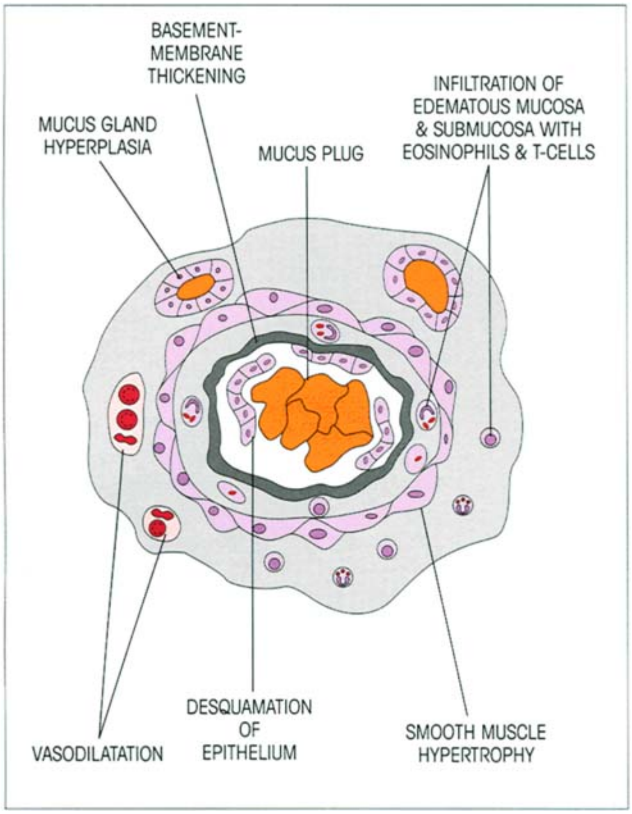


Figure 16.6. Pathological changes in asthma. Diagram of cross-section of an airway in severe asthma.

Figure 16.7. An atopic eczema reaction on the back of a knee of a child allergic to rice and eggs. (Kindly provided by Professor J. Brostoff.)

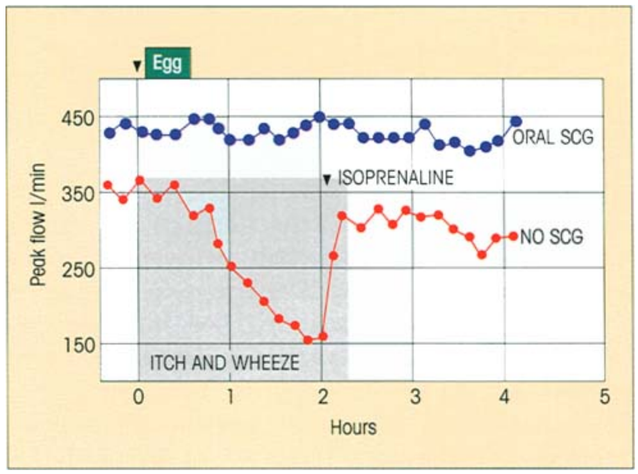


Figure 16.8. The role of gut sensitivity in the development of asthma to food allergens. A patient challenged by feeding with egg developed asthma within hours, as shown here by the depressed lung function test of measuring peak air flow; the symptoms at the end-organ stage were counteracted by the β-adrenoreceptor agonist, isoprenaline. However, oral sodium cromoglycate (SCG), which prevents antigen-specific mast cell triggering, also prevented the onset of asthma after oral challenge with egg. Note that SCG taken orally has no effect on the response of an asthmatic to inhaled allergen. (From Brostoff J. (1986) In Brostoff J. & Challacombe S.J. (eds) *Food Allergy*, p. 441. Baillière Tindall, London, reproduced with permission.)

food allergens in the gut has increased dramatically. Sensitization to egg white and cows' milk may even occur in early infancy through breast-feeding, with antigen passing into the mother's milk. Food additives such as sulfiting agents can also cause adverse reactions. Contact of the food with specific IgE on mast cells in the gastrointestinal tract may produce local reactions such as diarrhea and vomiting, or may allow the allergen to enter the body by causing a change in gut permeability through mediator release; the allergen may complex with antibodies and cause distal lesions by depositing in the joints, for example, or it may diffuse freely to other sensitized sites, such as the skin (figure 16.7) or lungs, where it will cause a further local anaphylactic reaction. Thus eating strawberries may produce urticarial reactions and egg may precipitate an asthmatic attack in appropriately sensitized individuals. The role of the sensitized gut in acting as a 'gate' to allow entry of allergens is strongly suggested by experiments in which oral sodium cromoglycate, a mast cell stabilizer, prevented subsequent asthma after ingestion of the provoking food (figure 16.8).

Anaphylactic drug allergy is manifest in the dramatic responses to drugs such as **penicillin** which haptenate body proteins by covalent coupling to induce IgE synthesis. In the case of penicillin, the β-lactam ring links to the ε-amino of lysine to form the penicilloyl determinant. The fine specificity of the IgE antibodies permits discrimination between closely similar drugs, such that some patients may be allergic to amoxicillin but tolerate benzylpenicillin which differs by only very minor modifications of the side-chains.

Pathologic mechanisms in asthma

We should now look in more depth at those events

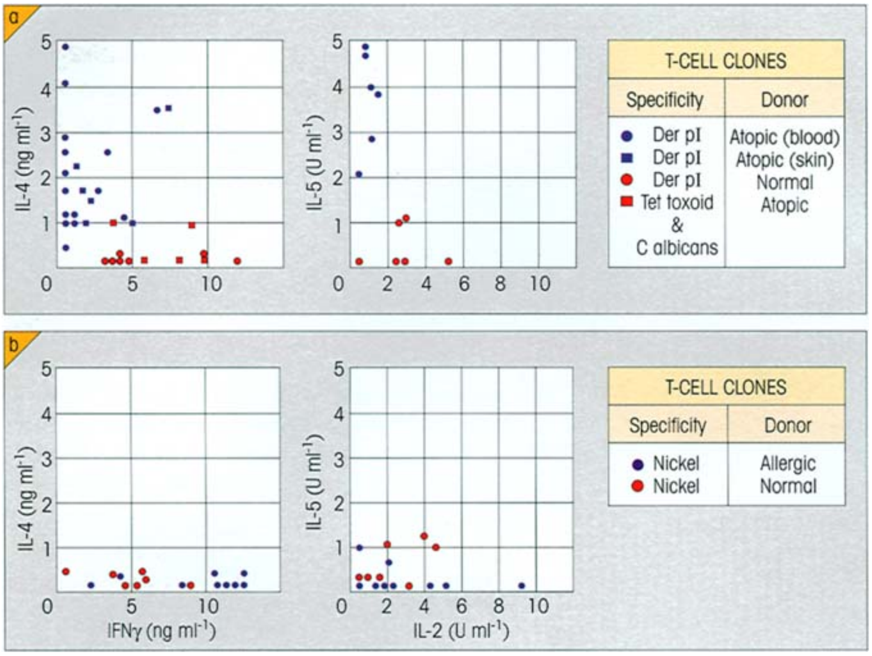


Figure 16.9. Th2 dominance in atopic allergy shown by cytokine profiles of antigen-specific CD4+ T-cell clones from (a) patients with type I atopic allergy and (b) subjects with type IV contact sensitivity, compared with normal controls. Each point represents the value for an individual clone. Archetypal Th1 clones have high IFNγ and IL-2 and low IL-4 and IL-5; Th2 clones show the converse. The high level of IL-4 drives the switch to IgE production by B-cells and further promotes the Th2 bias, which is reinforced by suppression of Th1 cells by NO· derived from cytokine-stimulated airway epithelial cells. (Data from Kapsenberg M.L., Wierenga E.A., Bos J.D. & Jansen H.M. (1991) *Immunology Today* **12**, 392.)

which generate the chronicity of asthma. Remember that there is an *early phase* bronchial response to inhaled antigen essentially involving mast cell mediators, and an inflammatory *late phase* dominated by eosinophils. **Both phases are IgE-dependent** as shown by their marked attenuation in the great majority of asthmatics treated for around 9 weeks with a humanized monoclonal antibody (RhuMAb-E25) specific for the binding site of human IgE to its high affinity receptor, which reduces IgE to almost undetectable levels and downregulates FcεRI expression. Activated mast cells make some contribution to eosinophil recruitment by secretion of an eosinophil chemoattractant and tryptase, which can switch on a protease-activated receptor (PAR-2) on the surface of endothelial and epithelial cells, leading to cytokine production and the expression of adhesion molecules which selectively recruit eosinophils and basophils. The most important trigger of the late phase reaction is now thought to be **activation of alveolar macrophages** through the interaction of allergen with IgE bound to the low affinity receptors (FcεRII; CD23), leading to a significant increase in the production of TNF and IL-1β. These cytokines then stimulate the release of the powerful **eosinophil chemoattractants**: eotaxin (CCL11), RANTES (CCL5) and MCP5 (CCL12) (cf. p. 187) from bronchial epithelial cells and fibroblasts. Note also that eotaxin and RANTES can contribute directly to local inflammation by IgE-independent degranulation of basophils.

A new player now enters the field: primed T-cells traffic into the inflamed site and are strongly attracted by eotaxin. Since the T-cell response is heavily skewed towards the **Th2 subset in asthma** (figure 16.9), encounter with the allergen on antigen-presenting cells will promote the synthesis of IL-4, -5 and -13. IL-4 stimulates further eotaxin release, while IL-5 upregulates chemokine receptors on eosinophils, maintains their survival through an inhibitory effect on natural apoptosis and is involved in their longer term recruitment from bone marrow.

Things now look bad for the bronchial tissues and a multitude of factors contribute to allergen-induced airway dysfunction: (i) a virtual soup of bronchoconstrictors, the leukotrienes being especially important, bathe the smooth muscle cells, (ii) edema of the airway wall, (iii) altered neural regulation of airway tone through binding of eosinophil major basic protein (MBP) to M2 autoreceptors on the nerve endings with increased release of acetylcholine, (iv) airway epithelial cell desquamation due to the toxic action of MBP, there being a strong correlation between the number of desquamated cells in bronchoalveolar lavage fluid and the concentration of MBP, (v) mucus hypersecretion due to IL-13 and, to a lesser extent, IL-4, leukotrienes and platelet activating factor acting on submucosal glands and their controlling neural elements, and finally (vi) a repair-type response involving the production of fibroblast growth factor, TGFβ and platelet-derived growth factor, the laying down of collagen, scar and fibrous tissue and hypertrophy of smooth muscle, leading to an exaggerated narrowing of the airways in response to a variety of environmental stimuli (figure 16.6). The wide range of cytokines and mediators produced by lung epithelial and endothelial cells, fibroblasts and smooth muscle cells may account for the persistence of airway inflammation and the permanent structural changes in chronic disease sufferers, even in the absence or apparent absence of ongoing exposure to inhalant allergens to which subjects are sensitized, a state where conventional immunotherapy might not be expected to be beneficial.

Unlike atopic asthmatics, **intrinsic asthmatics** have negative skin tests to common aeroallergens, no clinical or family history of allergy, normal levels of serum IgE and no detectable specific IgE antibodies to common allergens. Nonetheless, they resemble the atopics in important respects: bronchial biopsies show enhanced expression of IL-4, IL-13, RANTES and eotaxin, and of the mRNA for the ε germ line transcript and the ε heavy chain, suggestive of local IgE synthesis. Is there a role for virus-specific IgE or for IgE autoantibodies to the FcεRI?

The inflammatory infiltrate in **atopic dermatitis** closely resembles that in asthma. The epidermal dendritic cells have markedly upregulated FcεRI expression, and an incoming allergen will activate them either directly or as allergen–IgE complexes; the TNF, IL-1β and GM-CSF which they produce will, in turn, promote the release of the eosinophil chemoattractants, RANTES and eotaxin, from keratinocytes and fibroblasts. A newly identified keratinocyte CC chemokine, CTACK (CCL27; cf. p. 187), preferentially attracts skin-homing CLA$^+$ memory T-cells; these make up 80–90% of the T-cells in the infiltrate and account for the specific response to the offending allergen.

Etiological factors in the development of atopic allergy

There is a strong familial predisposition to the development of atopic allergy (figure 16.10). One factor is undoubtedly the overall ability to synthesize the IgE isotype—the higher the level of IgE in the blood, the greater the likelihood of becoming atopic (figure 16.10). Genetic studies show this to be linked to chromosome 13q. Linkages have also been established for other factors contributing to atopy: to chromosome 11q involving the FcεRIβ chain, to 2q containing the IL-1 cluster, and to 5q within the IL-4, -9, -13, GM-CSF and CD14 gene cluster. A C → T base change 159 bases upstream of the transcription start site for CD14, the high affinity receptor for bacterial LPS endotoxin, is significantly associated with high levels of soluble CD14 and low levels of IgE. What relevance might this have for atopic disease? Well, excessive exposure of babies to

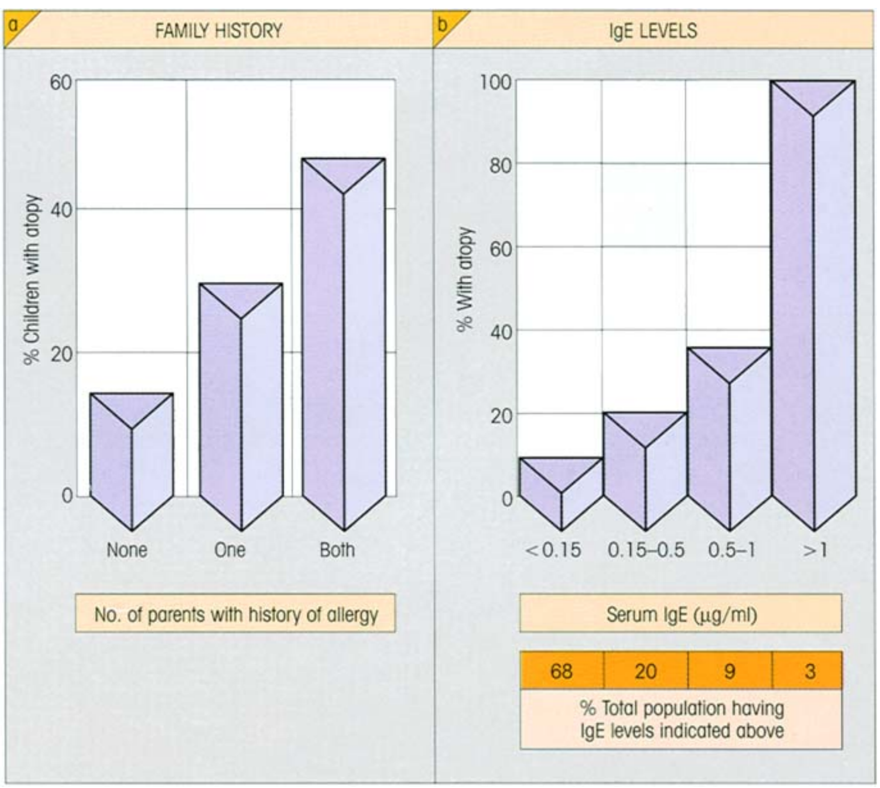


Figure 16.10. Risk factors in allergy: (a) family history; (b) IgE levels—the higher the serum IgE concentration, the greater the chance of developing atopy.

bacterial LPS in dirty homes, farms and households with dogs will tend to stimulate Th1 responses at the expense of Th2 and could account for the much lower incidence of atopy in these children. Indeed, a study of BCG vaccination in Japanese children revealed a strong inverse correlation between the development of delayed-type tuberculin reactions and the susceptibility to asthma. The current notion is that good hygiene in the early years is bad for you and is responsible for the steady increase in prevalence of atopy, the exposure to IL-12/Th1-inducing infections skewing the infant 'immunostat' more firmly towards the Th1 end of the spectrum and away from a Th2-related propensity to synthesize IgE. This may be oversimplified but it does make a good story—sorry, hypothesis. Other environmental influences, such as smoking and the use of beta-blockers during pregnancy, increase the incidence of atopy in the offspring, as does contact with highly allergenic proteins such as cows' milk, eggs, fish, nuts and Der p1 before mucosal protective mechanisms, especially IgA, are reasonably established. The old wives' tale that 'breast is best' would seem to have merit, although in some cases small amounts of dietary allergens may find their way into the mother's milk.

Clinical tests for allergy

Sensitivity is normally assessed by the response to intradermal challenge with antigen. The release of histamine and other mediators rapidly produces a **wheal and erythema** (figure 16.5), maximal within 30 minutes and then subsiding. These immediate wheal and flare reactions may be followed by a late phase reaction (cf. figure 16.5) which sometimes lasts for 24 hours, redolent of those seen following challenge of the bronchi and nasal mucosa of allergic subjects and similarly characterized by dense infiltration with eosinophils and T-cells.

The correlation between skin prick test responses and the **radioallergosorbent test** (RAST, see p. 113) for allergen-specific serum IgE is fairly good. In some instances, intranasal challenge with allergen may provoke a response even when both of these tests are negative, probably as a result of local synthesis of IgE antibodies.

The presence of proteins secreted from mast cells or eosinophils in the serum or urine could provide important surrogate markers of disease and might predict exacerbations.

Therapy

If one considers the sequence of reactions from initial exposure to allergen right through to the production of atopic disease, it can be seen that several points in the chain provide legitimate targets for therapy (figure 16.11).

Allergen avoidance. Avoidance of contact with *potential* allergens is often impractical, although, to give one example, feeding infants cows' milk at too early an age is discouraged. After sensitization, avoidance where possible is obviously worthwhile, but the reluctance of some parents to dispose of the family cat to stop little Algernon's wheezing is sometimes quite surprising.

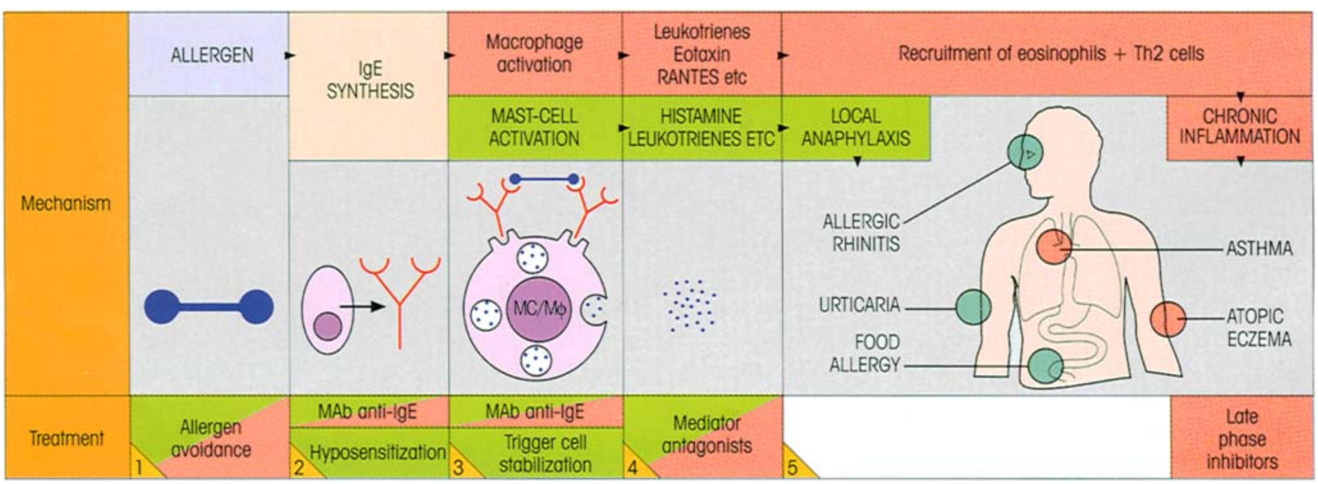


Figure 16.11. Atopic allergies and their treatment: sites of local responses and possible therapies. Events and treatments relating to local anaphylaxis in green and to chronic inflammation in red.

Modulation of the immunological response. Attempts to desensitize patients immunologically by repeated subcutaneous injection of small amounts of allergen have at least the merit of a long history and can lead to worthwhile improvement in individuals subject to insect venom anaphylaxis or hay fever, but are less effective in asthma. Injection of naked DNA coding for house dust mite allergen directly into the muscle, or oral administration of chitosan-coated plasmid DNA containing a dominant peanut allergen gene, reduced IgE synthesis and gave good protection against anaphylactic challenge in mice, but the long-term endogenous expression of allergen in sensitized humans might lead to precarious situations. The purpose of allergen hyposensitization therapy was originally to boost the synthesis of 'blocking' antibodies, whose function was to divert the allergen from contact with tissue-bound IgE. While this may well prove to be a contributory factor, downregulation of IgE synthesis by engagement of the FcγRIIB receptor (cf. p. 50) on B-cells by allergen-specific IgG linked to allergen molecules bound to surface IgE receptors also seems likely (cf. p. 202; see Chapter 11 on IgG regulation of Ab production). Additionally, if T-lymphocyte cooperation is important for IgE synthesis and eosinophil-mediated pathogenesis, the beneficial effects of antigen injection may also be mediated through induction of tolerant, anergic or suppressor T-cells. A switch from Th2 to Th1 by injection of heat-killed *Mycobacterium vaccae* or the administration of tolerizing or antagonist peptide epitopes represent other possible therapeutic modalities. Fortunately, most patients respond to a remarkably limited number of T-cell epitopes on any given allergen, and so it may not be necessary to tailor the therapeutic peptide to each individual. Clinical trials with high doses of Fel d 1-derived peptides from cat allergen have resulted in decreased sensitivity, although isolated late responses to allergen by direct T-cell activation may be induced. A case can be made for future prophylactic hyposensitization of children with two asthmatic parents who have at least a 50% probability of developing the disease.

The humanized anti-IgE monoclonal, RhuMAb-E25, directed against the FcεRI-binding domain of IgE (cf. p. 324) may provide an exciting new therapy for severe forms of asthma and atopic dermatitis, because it impacts the disease process at several points. It reduces the circulating IgE levels almost to vanishing point by direct neutralization, and decreases IgE synthesis through binding to the inhibitory FcγRIIB as it cross-links the surface immunoglobulin on IgE-producing B-cells (cf. the effect of allergen-specific IgG mentioned above).

Stabilization of the triggering cells. This precipitous fall in circulating IgE caused by RhuMAb-E25 decreases the occupancy of mast cell FcεRI receptors, and reduces receptor synthesis, perhaps through the lack of positive feedback mediated by the β chains; an additional nudge to the downregulation of receptor biosynthesis may be provided by cross-linking of the free Fcε-binding domain on the mast cell surface (cf. figure 16.1) and subsequent engagement of the FcγRIIB. Signaling through allergen will obviously be far less effective because only a small number of receptors per cell will be activated. Last and by no means least, similar mechanisms will operate to limit the allergen-induced activation of macrophages and hence eosinophil recruitment. At the drug level, much relief has been obtained with agents such as inhalant isoprenaline and **sodium cromoglycate**, a member of the chromone family, which render mast cells resistant to triggering. Sodium cromoglycate blocks chloride channel activity and maintains cells in a normal resting physiological state, which probably accounts for its inhibitory effects on a wide range of cellular functions, such as mast cell degranulation, eosinophil and neutrophil chemotaxis and mediator release, and reflex bronchoconstriction. Some or all of these effects are responsible for its anti-asthmatic actions. Strangely, it also inhibits the release of IgE from tonsils of nonatopic subjects stimulated with IL-4.

The triggering of macrophages through allergen interaction with surface-bound IgE is clearly a major initiating factor for late reactions, as discussed above, and resistance to this stimulus can be very effectively achieved with corticosteroids. Unquestionably, **inhaled corticosteroids** have revolutionized the treatment of asthma. Their principal action is to suppress the transcription of multiple inflammatory genes, including in the present context those encoding cytokine production.

Mediator antagonism. **Histamine H_1-receptor antagonists** have for long proved helpful in the symptomatic treatment of atopic disease. Newer drugs such as loratadine and fexofenadine are effective in rhinitis and in reducing the itch in atopic dermatitis, although they have little benefit in asthma. Cetirizine additionally has useful effects on eosinophil recruitment in the late phase reaction. An important recent advance has been the introduction of long-acting inhaled **β_2-agonists** such as salmeterol and formoterol which are bronchodilators and protect against bronchoconstriction for over 12 hours. Potent **leukotriene antagonists** such as Pranlukast also block constrictor challenges and show striking efficacy in certain patients, particularly aspirin-sensitive asthmatics (logical if you think about it). At the experimental level, an antibody to IL-5 had an effect on an allergy model in primates lasting several months.

Theophylline was introduced for the treatment of asthma more than 50 years ago and remains the single most prescribed drug for asthma worldwide. As a **phosphodiesterase (PDE) inhibitor** it increases intracellular cAMP, thereby causing bronchodilatation, inhibition of IL-5-induced prolongation of eosinophil survival and probably suppression of eosinophil migration into the bronchial mucosa. Good news for the patient. There are many isoenzyme forms of PDE, and bronchodilators such as benafentrine are currently being developed.

Attacking chronic inflammation. Certain drugs impede atopic disease at more than one stage. **Cetirizine** is a case in point with its dual effects on the histamine receptor and on eosinophil recruitment. **Corticosteroids** seem to do almost everything; apart from their role in stabilizing macrophages, they solidly inhibit the activation and proliferation of the Th2 cells, which are the dominant underlying driving force in chronic asthma, and may call a halt to the development of irreversible narrowing of the airways. So it is that new generation inhaled steroids (e.g. budesonide, mometasone furoate, fluticasone propionate) with high anti-inflammatory potency but minimal side-effects due to hepatic metabolism, provide first-line therapy for most chronic asthmatics, with supplementation by long-acting β_2-agonists and theophylline.

ANTIBODY-DEPENDENT CYTOTOXIC HYPERSENSITIVITY (TYPE II)

Where an antigen is present on the surface of a cell, combination with antibody will encourage the demise of that cell by promoting contact with phagocytes, either by reduction in surface charge, or by **opsonic adherence** to Fcγ or C3b receptors. Cell death may also occur through activation of the full **complement** system up to C8 and C9 producing **direct membrane damage** (figure 16.12). Although, in the case of hemolytic antibodies, the generation of a single active complement site is enough to cause erythrocyte lysis, other cells appear to have repair mechanisms and it is likely that several complement sites need to be recruited in order to overwhelm the cell's defenses.

The operation of a quite distinct cytotoxic mechanism derives from the finding that target cells coated with low concentrations of IgG antibody can be killed 'nonspecifically' through an extracellular nonphagocytic mechanism involving nonsensitized leukocytes which bind to the target by their specific receptors for the Cγ2 and Cγ3 domains of IgG Fc (figures 16.12 and 16.13). It should be noted that this so-called **antibody-dependent cellular cytotoxicity (ADCC)** may be exhibited by both phagocytic and nonphagocytic myeloid cells (polymorphs and monocytes) and by large granular lymphocytes with Fc receptors dubbed 'K-cells', which are almost certainly identical with the natural killer (NK) cells (see p. 32). Contact between the effector and target cells is essential and activity is inhibited by cytochalasin B which interferes with cell

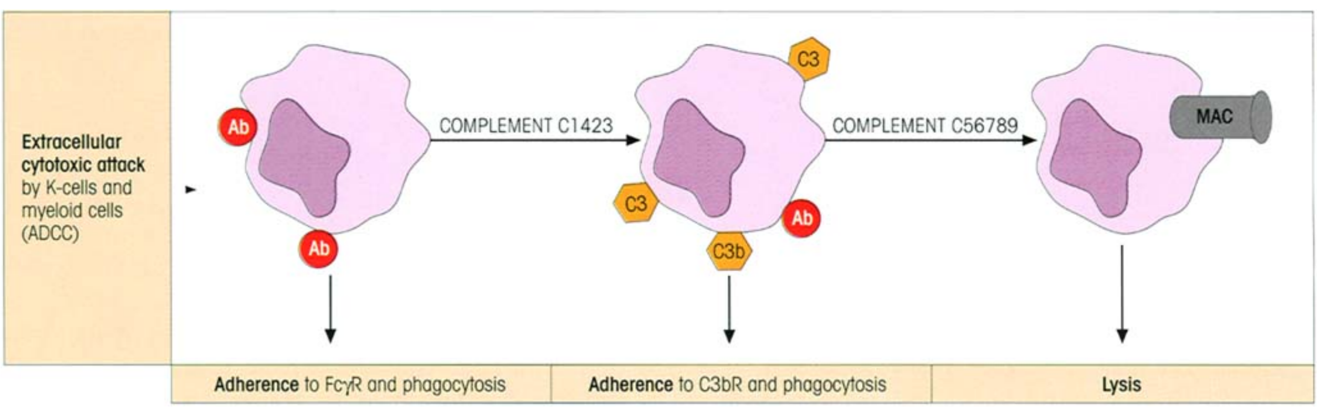


Figure 16.12. Antibody-dependent cytotoxic hypersensitivity (type II). Antibodies directed against cell surface antigens cause cell death not only by C-dependent lysis but also by Fcγ and C3b adherence reactions leading to phagocytosis, or through nonphagocytic extracellular killing by certain lymphoid and myeloid cells (antibody-dependent cellular cytotoxicity). IgG cytotoxic antibodies directed to tumors or to glomerular basement membrane are 10–100 times more pathogenic in animals deficient in FcγRIIB relative to their wild-type controls, pointing to a balance of activating and inhibitory signals controlling the dominant effector pathway in these responses, and implicating the macrophage as the critical cell type involved.

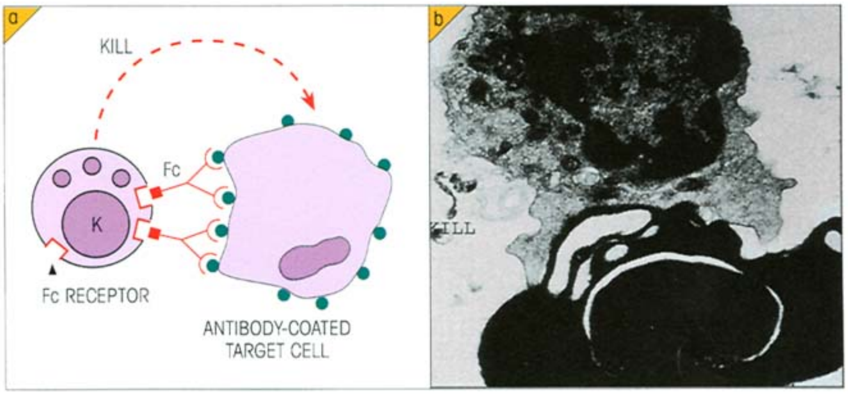


Figure 16.13. Killing of Ab-coated target by antibody-dependent cellular cytotoxicity (ADCC). Fcγ receptors bind the effector to the target which is killed by an extracellular mechanism. Human monocytes and IFNγ-activated neutrophils kill Ab-coated tumor cells using their FcγRI receptors; lymphocytes (NK cells) kill hybridoma targets through FcγRIII receptors. (a) Diagram of effector and target cells. (b) Electron micrograph of attack on Ab-coated chick red cell by a mouse large granular lymphocyte showing close apposition of effector and target and vacuolation in the cytoplasm of the latter. ((b) Courtesy of P. Penfold.)

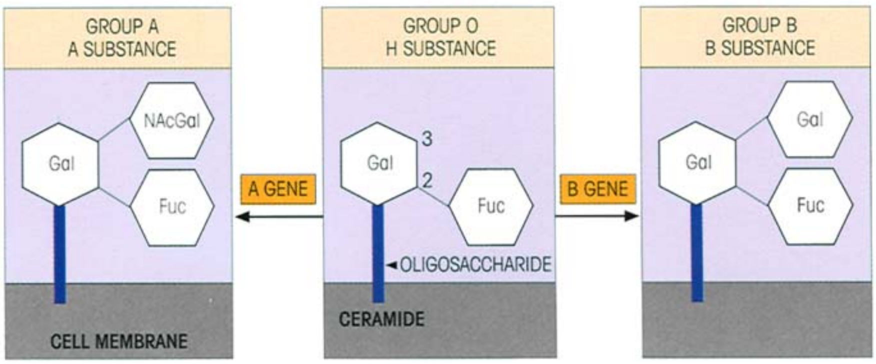


Figure 16.14. The ABO system. The allelic genes A and B code for transferases which add either *N*-acetylgalactosamine (NAcGal) or galactose (Gal), respectively, to H substance (Fuc, fucose). The oligosaccharide is anchored to the cell membrane by coupling to a sphingomyelin called ceramide. Eighty-five per cent of the population secrete blood group substances in the saliva, where the oligosaccharides are present as soluble polypeptide conjugates formed under the action of a secretor (*se*) gene.

movement, and by aggregated IgG which binds firmly to the Fc receptors and blocks their ability to interact with antibody on the surface of the target.

ADCC can be readily observed as a phenomenon *in vitro*; to give examples, human K-cells have been shown to be strikingly unpleasant to chicken red cells coated with rabbit antibody, while schistosomules coated with either IgG or IgE can be killed by eosinophils (cf. figure 13.22). Whether ADCC plays a positive role *in vivo* remains a tricky question, but functionally this extracellular cytotoxic mechanism would be expected to be of significance where the target is too large for ingestion by phagocytosis, e.g. large parasites and solid tumors. It could also act as a back-up system for T-cell killing.

Type II reactions between members of the same species (alloimmune)

Transfusion reactions

Of the many different polymorphic constituents of the human red cell membrane, **ABO blood groups** form the dominant system. The antigenic groups A and B are derived from H substance (figure 16.14) by the action of glycosyltransferases encoded by *A* or *B* genes, respectively. Individuals with both genes (group AB) have the two antigens on their red cells, while those lacking these genes (group O) synthesize H substance only. Antibodies to A or B occur spontaneously when the antigen is absent from the red cell surface; thus a person of blood group A will possess anti-B and so on. These **isohemagglutinins** are usually IgM and probably belong to the class of 'natural antibodies'; they would be boosted through contact with antigens of the gut flora which are structurally similar to the blood group carbohydrates, so that the antibodies formed cross-react with the appropriate red cell type. If an individual is blood group A, he/she would be tolerant to antigens closely similar to A and would only form cross-reacting antibodies capable of agglutinating B red cells; similarly an O individual would make anti-A and anti-B (table 16.2). On transfusion, mismatched red cells will be coated by the isohemagglutinins and will cause severe complement-mediated intravascular hemolysis.

Clinical refractoriness to platelet transfusions is frequently due to HLA alloimmunization, but one can usually circumvent this problem by depleting the platelets of leukocytes.

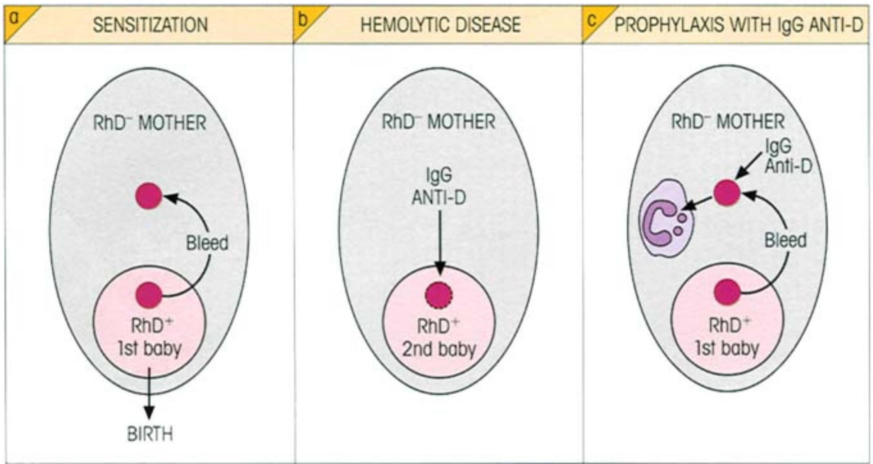


Figure 16.15. Hemolytic disease of the newborn due to rhesus incompatibility. (a) RhD+ve red cells from the first baby sensitize the RhD–ve mother. (b) The mother's IgG anti-D crosses the placenta and coats the erythrocytes of the second RhD+ve baby causing type II hypersensitivity hemolytic disease. (c) IgG anti-D given prophylactically at the first birth removes the baby's red cells through phagocytosis and prevents sensitization of the mother.

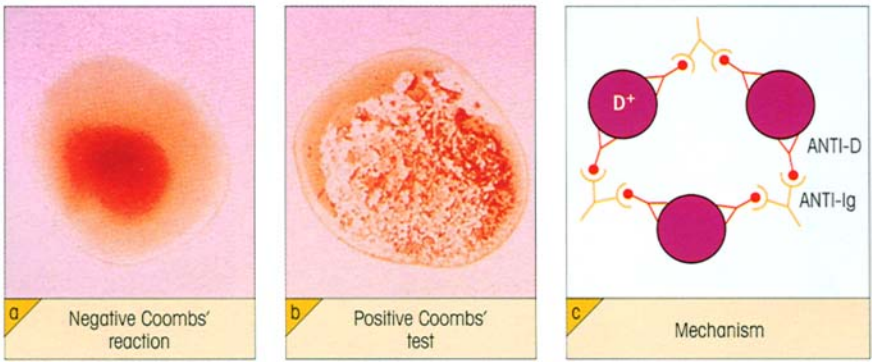


Figure 16.16. The Coombs' test for antibody-coated red cells used for detecting rhesus antibodies and in the diagnosis of autoimmune hemolytic anemia (cf. table 19.2, Note 5, p. 402). (Photographs courtesy of Professor A. Cooke.)

Table 16.2. ABO blood groups and serum antibodies.

BLOOD GROUP (PHENOTYPE)	GENOTYPE	ANTIGEN	SERUM ANTIBODY
A	*AA,AO*	A	ANTI-B
B	*BB,BO*	B	ANTI-A
AB	*AB*	A and B	NONE
O	*OO*	H	ANTI-A ANTI-B

Rhesus incompatibility

The **rhesus (Rh) blood groups** form the other major antigenic system, the RhD antigen being of the most consequence for isoimmune reactions. A mother with an RhD–ve blood group (i.e. *dd* genotype) can readily be sensitized by red cells from a baby carrying RhD antigens (*DD* or *Dd* genotype). This occurs most often at the birth of the first child when a placental bleed can release a large number of the baby's erythrocytes into the mother. The antibodies formed are predominantly of the IgG class and are able to cross the placenta in any subsequent pregnancy. Reaction with the D-antigen on the fetal red cells leads to their destruction through opsonic adherence, giving hemolytic disease of the newborn (figure 16.15).

These anti-D antibodies fail to agglutinate RhD+ve red cells *in vitro* ('incomplete antibodies') because the low density of antigenic sites does not allow sufficient antibody bridges to be formed between the negatively charged erythrocytes to overcome the electrostatic repulsive forces. Erythrocytes coated with anti-D can be made to agglutinate by addition of albumin or of an anti-immunoglobulin serum (Coombs' reagent; figure 16.16).

If a mother has natural isohemagglutinins which can react with any fetal erythrocytes reaching her circulation, sensitization to the D-antigens is less likely due to 'deviation' of the red cells away from the antigen-sensitive cells. For example, a group O RhD–ve mother with a group A RhD+ve baby would destroy any fetal erythrocytes with her anti-A before they could immunize to produce anti-D. In an extension of this principle, **RhD–ve mothers are now treated prophylactically** with small amounts of avid IgG anti-D at the time of birth of the first child, and this greatly reduces the risk of sensitization. Another success for immunology.

Another example of disease resulting from transplacental passage of maternal antibodies is **neonatal alloimmune thrombocytopenia**. The fall in platelet numbers is greatly ameliorated by high-dose i.v. injections of pooled human IgG, thought by some to involve anti-idiotype networks (cf. p. 443), although the efficacy of Fcγ fragments, anti-RhD and anti-FcγR rather point the finger at blockade of the Fcγ receptors.

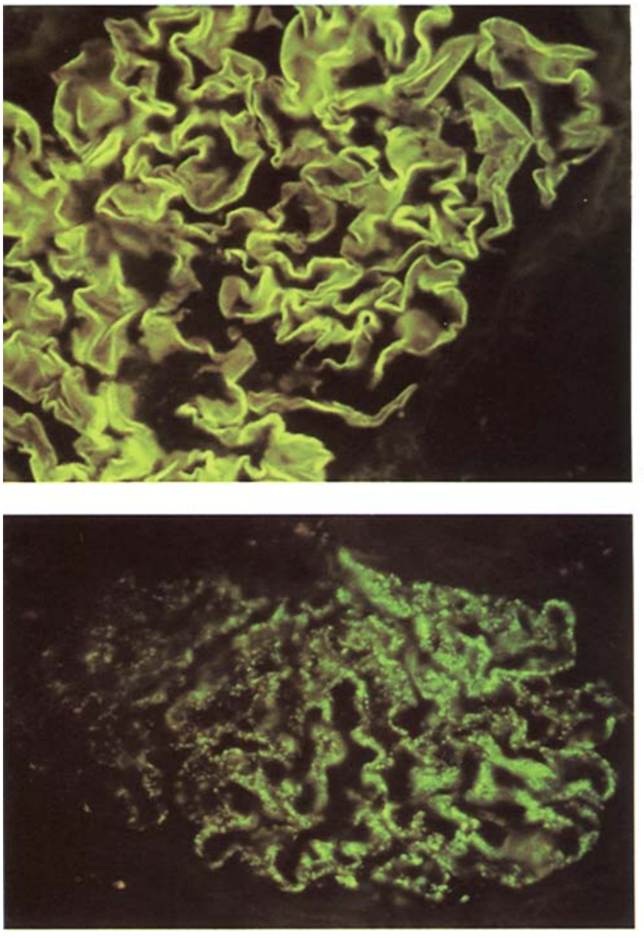
Organ transplants

A long-standing allograft which has withstood the first onslaught of the cell-mediated reaction can evoke humoral antibodies in the host directed against surface transplantation antigens on the graft. These may be directly cytotoxic or cause adherence of phagocytic cells or 'nonspecific' attack by K-cells. They may also lead to platelet adherence when they combine with antigens on the surface of the vascular endothelium (figure 17.6, p. 354). Hyperacute rejection is mediated by preformed antibodies in the graft recipient.

Autoimmune type II hypersensitivity reactions

Autoantibodies to the patient's own red cells are produced in **autoimmune hemolytic anemia**. They react at 37°C with epitopes on antigens of the rhesus complex distinct from those which incite transfusion reactions. Red cells coated with these antibodies have a shortened half-life, largely through their adherence to phagocytic cells in the spleen. Similar mechanisms account for the anemia in patients with cold hemagglutinin disease who have monoclonal anti-I after infection with *Mycoplasma pneumoniae*, and in some cases of paroxysmal cold hemoglobinuria associated with the actively lytic Donath–Landsteiner antibodies specific for blood group P. These antibodies are primarily of IgM isotype and only react at temperatures well below 37°C. IgG platelet autoantibodies are responsible for the depletion of platelets in **idiopathic thrombocytopenic purpura**; although the relative contributions of C3b and Fcγ receptors to their clearance by phagocytosis is still a matter of dispute, the persistence of opsonized platelets when Fcγ receptors are blocked (see above) or deficient (in a mouse model) does argue strongly that Fcγ interactions with its receptor are pivotal.

The serums of patients with Hashimoto's thyroiditis contain antibodies which, in the presence of complement, are directly cytotoxic for isolated human thyroid cells in culture. In Goodpasture's syndrome (included here for convenience), antibodies to kidney glomerular basement membrane are present. Biopsies show



(a)

(b)

Figure 16.17. Glomerulonephritis: (a) due to linear deposition of antibody to glomerular basement membrane, here visualized by staining the human kidney biopsy with a fluorescent anti-IgG; and (b) due to deposition of antigen–antibody complexes, which can be seen as discrete masses lining the glomerular basement membrane following immunofluorescent staining with anti-IgG. Similar patterns to these are obtained with a fluorescent anti-C3. (Courtesy of Dr S. Thiru.)

these antibodies, together with complement components, bound to the basement membranes where the action of the full complement system leads to serious damage (figure 16.17). One could also include the stripping of acetylcholine receptors from the muscle endplate by autoantibodies in myasthenia gravis as a further example of type II hypersensitivity.

Type II drug reactions

This is complicated. Drugs may become coupled to body components and thereby undergo conversion from a hapten to a full antigen which will sensitize certain individuals (we don't know which). If IgE antibodies are produced, anaphylactic reactions can result. In some circumstances, particularly with topically applied ointments, cell-mediated hypersensitivity may be induced. In other cases where coupling to serum proteins occurs, the possibility of type III complex-mediated reactions may arise. In the present context, we are concerned with those instances in which the drug appears to form an antigenic complex with the surface of a formed element of the blood and evokes the production of antibodies which are cytotoxic for the cell–drug complex. When the drug is withdrawn, the sensitivity is no longer evident. Examples of this mechanism have been seen in the **hemolytic anemia** sometimes associated with continued administration of chlorpromazine or phenacetin, in the **agranulocytosis** associated with the taking of amidopyrine or of quinidine, and the now classic situation of **thrombocytopenic purpura** which may be produced by sedormid, a sedative of yesteryear. In the latter case, freshly drawn serum from the patient will lyse platelets in the presence, but not in the absence, of sedormid; inactivation of complement by preheating the serum at 56°C for 30 minutes abrogates this effect.

IMMUNE COMPLEX-MEDIATED HYPERSENSITIVITY (TYPE III)

The body may be exposed to an excess of antigen over a protracted period in a number of circumstances: persistent infection with a microbial organism, autoimmunity to self-components and repeated contact with environmental agents. The union of such antigens and antibodies to form a complex within the body may well give rise to acute inflammatory reactions through a variety of mechanisms (figure 16.18). For a start, complexes can stimulate macrophages through their Fcγ-receptors to generate the release of proinflammatory cytokines IL-1 and TNF, reactive oxygen intermediates and nitric oxide (figure 16.18). Complexes which are insoluble often cannot be digested after phagocytosis by macrophages and so provide a persistent activating stimulus. If complement is fixed, anaphylatoxins will be released as split products of C3 and C5 and these will cause release of mast cell mediators with vascular permeability changes. The chemotactic factors also produced will lead to an influx of polymorphonuclear leukocytes which begin the phagocytosis of the immune complexes; this in turn results in the extracellular release of the neutrophil granule contents, particularly when the complex is deposited on a basement membrane and cannot be phagocytosed (so-called 'frustrated phagocytosis'). The proteolytic enzymes (including neutral proteinases and collagenase), kinin-forming enzymes, polycationic proteins and reactive oxygen and nitrogen intermediates which are released will of course damage local tissues and intensify the inflammatory responses. Further havoc may be mediated by reactive lysis in which activated C5,6,7 becomes adventitiously attached to the surface of nearby cells and binds C8,9. Intravascular complexes can aggregate platelets with two consequences: they provide yet a further source of vasoactive amines and may also form microthrombi which can lead to local ischemia. (The discerning reader will appreciate the need for the system of inhibitors present in the body.)

The outcome of the formation of immune complexes *in vivo* depends not only on the absolute amounts of antigen and antibody, which determine the intensity of the reaction, but also on their *relative* proportions, which govern the nature of the complexes (cf. figure 6.2, p. 110) and hence their distribution within the body. Between **antibody excess** and **mild antigen excess**, the complexes are rapidly precipitated and tend to be localized to the site of introduction of antigen, whereas in **moderate** to **gross antigen excess**, soluble complexes are formed.

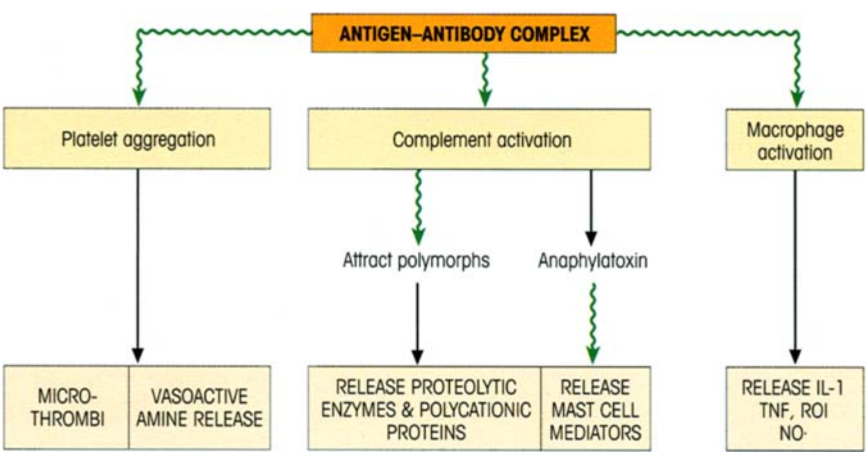


Figure 16.18. Immune complex-mediated (type III) hypersensitivity—underlying pathogenic mechanisms.

Covalent attachment of C3b prevents the Fc–Fc interactions required to form large insoluble aggregates, and these small complexes bind to CR1 complement receptors on the human erythrocyte and are transported to fixed macrophages in the liver where they are safely inactivated. If there are defects in this system, for example deficiencies in classical pathway components, or perhaps if the system is overloaded, then widespread disease involving deposition in the kidneys, joints and skin may result.

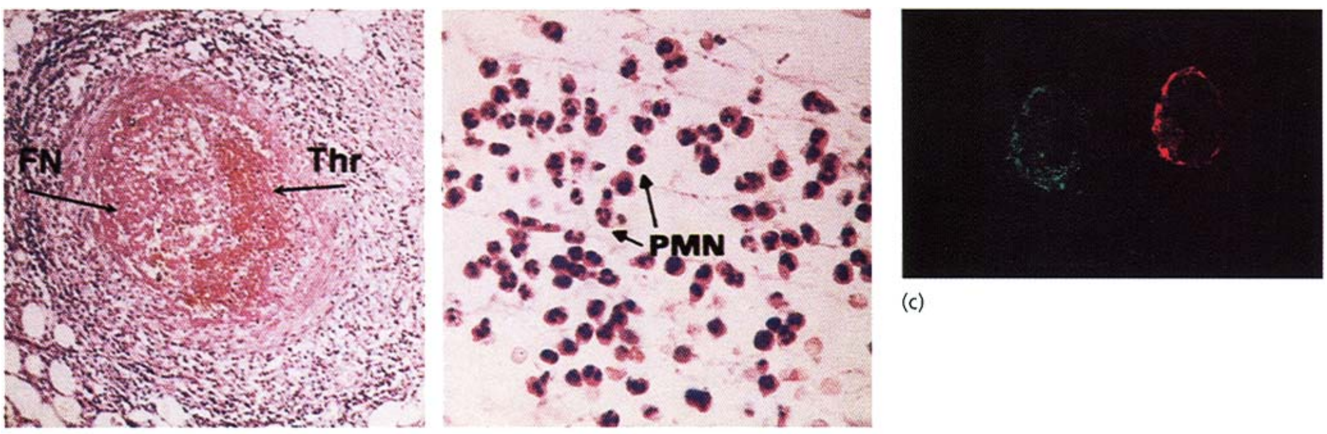
Inflammatory lesions due to locally formed complexes

The Arthus reaction

Maurice Arthus found that injection of soluble antigen intradermally into hyperimmunized rabbits with high levels of precipitating antibody produced an erythematous and edematous reaction reaching a peak at 3–8 hours which then usually resolved. The lesion was characterized by an intense infiltration with neutrophils (cf. figure 16.19a and b). The injected antigen precipitates with antibody often within the venule, too fast for the classical complement system to prevent it; subsequently, the complex binds complement and, using fluorescent reagents, antigen, immunoglobulin and complement components can all be demonstrated in this lesion, as illustrated by the inflammatory response to deposits of immune complexes containing hepatitis B surface antigen in a patient with periarteritis nodosa (figure 16.19c). Anaphylatoxin production, mast cell degranulation, macrophage activation, platelet aggregation and influx of neutrophils all make their contribution. The Arthus reaction can be attenuated by depletion of neutrophils by nitrogen mustard or of complement by anti-C5a; soluble forms of the complement regulatory proteins CD46 (membrane cofactor protein) and CD55 (delay accelerating factor) are also inhibitory.

Reactions to inhaled antigens

Intrapulmonary Arthus-type reactions to exogenous inhaled antigen appear to be responsible for a number of hypersensitivity disorders in humans. The severe respiratory difficulties associated with **farmer's lung** occur within 6–8 hours of exposure to the dust from mouldy hay. The patients are found to be sensitized to thermophilic actinomycetes which grow in the mouldy hay, and extracts of these organisms give precipitin reactions with the subject's serum and Arthus reactions on intradermal injection. Inhalation of bacterial spores present in dust from the hay introduces antigen into the lungs and a complex-mediated hypersensitivity reaction occurs. Similar situations arise in pigeon-fancier's disease, where the antigen is probably serum protein present in the dust from dried feces, in rat handlers sensitized to rat serum proteins excreted in the urine (figure 16.20) and in many other



(a) (b) (c)

Figure 16.19. Histology of acute inflammatory reaction in polyarteritis nodosa associated with immune complex formation with hepatitis B surface (HBs) antigen. (a) A vessel showing thrombus (Thr) formation and fibrinoid necrosis (FN) is surrounded by a mixed inflammatory infiltrate, largely polymorphs. (b) High-power view of acute inflammatory response in loose connective tissue of patient with polyarteritis nodosa—polymorphs (PMN) are prominent. (c) Immunofluorescence studies of immune complexes in the renal artery of a patient with chronic hepatitis B infection stained with fluoresceinated antihepatitis B antigen (left) and rhodaminated anti-IgM (right). The presence of both antigen and antibody in the intima and media of the arterial wall indicates the deposition of the complexes at this site. IgG and C3 deposits are also detectable with the same distribution. ((a) and (b) provided by courtesy of Professor N. Woolf; (c) kindly provided by Professor A. Nowoslowski.)